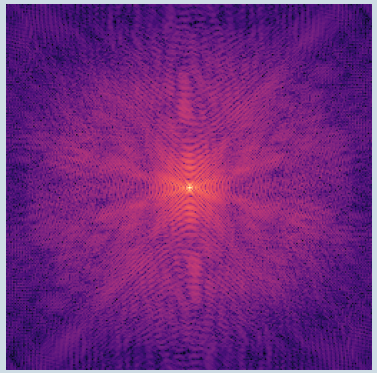
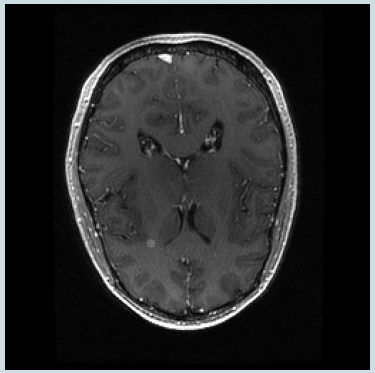
AI for Medical Imaging Science

From Physical Measurement to Clinical and Scientific Knowledge

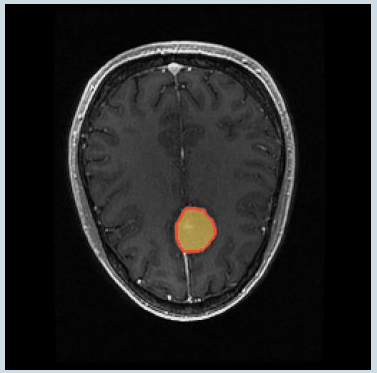
Essam Rashed, Ph.D.
Graduate School of Information Science,
University of Hyogo



Measurements (*k*-space)



Reconstructed image



Quantitative map

Lesion volume
16.3 cm³ (15.5–17.1)
Enhancement
solid, T1c
Evidence conf.
0.81 (calibrated)

Measurements ± uncertainty

FINDINGS
Enhancing parafalcine lesion…
IMPRESSION
Provisional — requires radiologist review

Report & research data

Learning Objectives and Roadmap

From physical measurement to evidence-grounded clinical reporting in three hours

1. Relate physical acquisition to computational reconstruction
2. Formulate imaging as forward and inverse problems
3. Explain AI architectures for reconstruction, segmentation, classification, quantification
4. Compare CNNs, Transformers, self-supervised and foundation models
5. Explain why medical foundation models need modality-specific, clinically aware adaptation
6. Understand multimodal learning across sequences, modalities, text, clinical data
7. Explain how vision–language models generate clinical reports
8. Design an end-to-end pipeline from raw data to a structured report
9. Evaluate accuracy, uncertainty, robustness, validity, and safety
10. Explain opportunities and limits of agentic AI in imaging
11. Identify research problems for master's or doctoral projects
12. Treat image-derived quantities as measurements with bias, precision, and uncertainty

Module I

Physics, quantitative measurement, inverse problems across sciences, CNN / U-Net / Transformers, evaluation

Module II

Self-supervision, foundation models, SAM family, diffusion, VLMs, report generation

Case studies 3–5

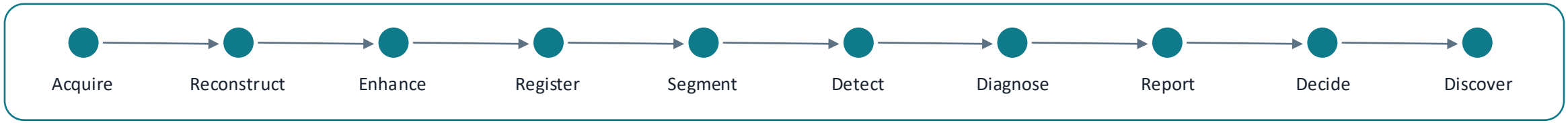
Module III

Agentic AI, uncertainty, verification, hybrid models, scientific discovery

Case study 6 · growth-model identifiability

Module I — Foundations of AI for Medical Imaging Science

Imaging physics · quantitative measurement · inverse problems across sciences · CNNs, U-Net, Transformers · clinical tasks · evaluation



Landmark Papers 2023–2026: A Map



Where recent top-journal work sits on the pipeline — and how strong its evidence is

Paper	Venue	Stage	Main contribution	Evidence type
Whole-body MRI at 0.05 T	Science 2024	Acquire · reconstruct	DL interference cancellation + image formation	Technical feasibility, human imaging
PANDA (pancreatic cancer)	Nat. Med. 2023	Detect	Cancer detection on non-contrast CT	Multicentre + real-world retrospective
RETFound	Nature 2023	Represent	Retinal foundation model; ‘oculomics’	Retrospective incl. external
CHIEF · Prov-GigaPath · UNI · Virchow	Nature / Nat. Med. 2024	Represent	Pathology foundation models	Retrospective, many cohorts
CT-FM (cancer imaging biomarkers)	Nat. Mach. Intell. 2024	Quantify	Self-supervised biomarker features	Retrospective incl. external
MedSAM · BiomedParse	Nat. Commun. 2024 · Nat. Methods 2025	Segment	Promptable / text-prompted segmentation	Retrospective benchmarks
CONCH · PathChat · EchoCLIP	Nat. Med. 2024 · Nature 2024	Interpret	Vision–language beyond radiology	Benchmarks + expert evaluation
RadFM · CT-CLIP/CT-CHAT · BrainGPT	Nat. Commun. 2025 · Nat. BME 2026	Interpret · report	Generalist and 3D radiology VLMs	Benchmarks + expert evaluation
Flamingo-CXR	Nat. Med. 2025	Report	Clinician–AI report collaboration	Expert reader evaluation
MASAI · PRAIM (mammography)	Lancet Oncol. 2023 · Nat. Med. 2025	Decide	AI-supported screening	RCT · nationwide real-world

Details on later slides. Compiled from the lecturer’s reading without live database access — verify bibliographic details before distribution.

Why AI for Medical Imaging?

Cross-cutting

Medical imaging is a measurement science; AI is a new way to solve its inference problems

Data complexity

- 3D/4D volumes, multiple sequences and modalities
- Longitudinal studies, reports, labs, genomics
- Scanner- and protocol-dependent signal

Limits of classical computation

- Hand-crafted priors (smoothness, sparsity)
- Slow iterative solvers at clinical scale
- Feature engineering per task

Clinical bottlenecks

- Expensive, variable expert annotation
- Inter-reader disagreement
- Growing reporting workload and fatigue

AI for science, not just recognition

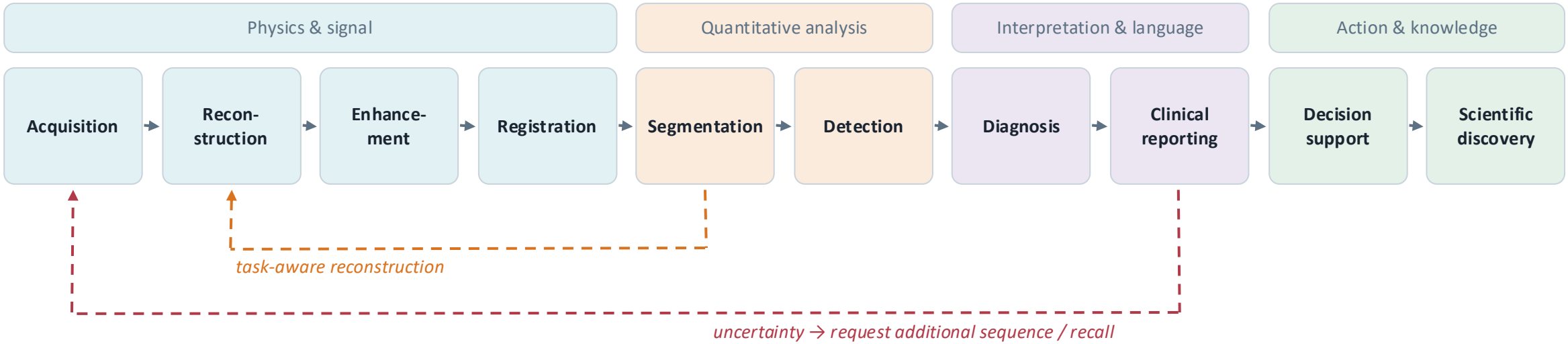
- Infer physical state from indirect measurements
- Quantify uncertainty and evidence
- Generate testable hypotheses

Two endpoints run through this lecture: a verifiable clinical statement and a reproducible scientific measurement. When does a model produce knowledge we can verify, rather than a pattern we cannot check?

An Integrated View of Medical Imaging AI

End-to-end pipeline

Every stage inherits errors and uncertainty from the stages upstream



Error propagation

A misregistration of a few mm between FLAIR and T1c propagates into every voxel-wise multi-sequence decision downstream.

Information bottleneck

Information discarded at acquisition or reconstruction cannot be recovered by any downstream model.

Closing the loop

Module III asks whether downstream uncertainty can steer acquisition, reconstruction, and reporting.

Imaging Modalities as Measurement Systems

Each modality measures a different physical quantity through a different forward operator

Modality	Physical quantity	Measurement	Noise model	Reconstruction	Typical clinical use
MRI	Proton magnetisation, relaxation (T1, T2)	k-space (Fourier) samples	Complex Gaussian	Inverse FFT, SENSE, CS, learned	Neuro, MSK, oncology, cardiac
CT	Linear attenuation μ(x)	Projection line integrals	≈ Poisson in photon counts	FBP, iterative, learned	Trauma, chest, abdomen, stroke
Ultrasound	Acoustic impedance, scattering	Echo RF signals per element	Multiplicative speckle	Delay-and-sum beamforming	Obstetrics, cardiac, abdomen
PET	Radiotracer concentration	Coincidence counts per LOR	Poisson emission	MLEM / OSEM	Oncology staging, neuro, cardiac

Common abstraction for the whole module: $y = A(x) + \varepsilon$ — known (or partly known) physics A, unknown object x, structured noise ε.

MRI Physics: Magnetisation and Relaxation

Image contrast is engineered by timing (TR, TE, TI) relative to tissue relaxation times

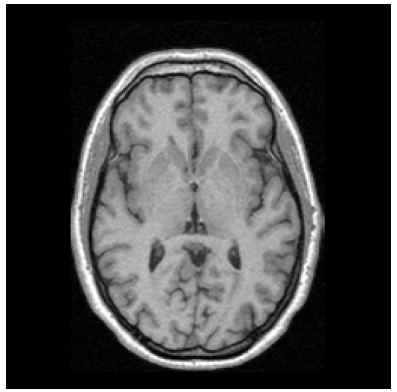
relaxation after a 90° pulse

$$M_z(t) = M_0\,(1 - e^{-t/T_1}) \qquad M_{xy}(t) = M_0\,e^{-t/T_2}$$

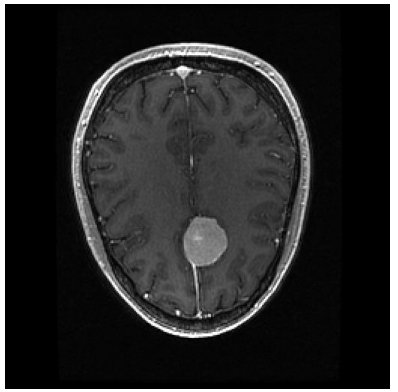
simplified spin-echo signal

$$S \approx PD \cdot (1 - e^{-TR/T_1}) \cdot e^{-TE/T_2}$$

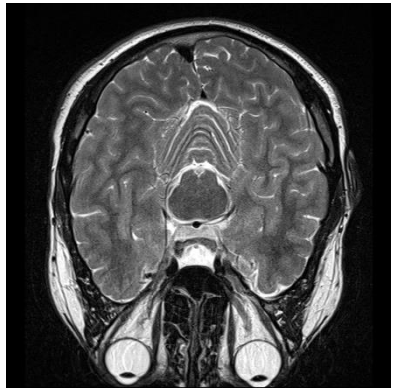
- **Larmor precession:** $\omega_0 = \gamma B_0$; RF excitation tips M into the transverse plane
- **T1-weighted** (short TR, short TE): fat bright, CSF dark, anatomy
- **T2-weighted** (long TR, long TE): fluid and oedema bright
- **FLAIR:** inversion with $TI \approx T_1,\text{CSF} \cdot \ln 2$ nulls CSF; oedema/infiltration bright
- **T1 + contrast (T1c):** gadolinium shortens T1 where the blood–brain barrier leaks



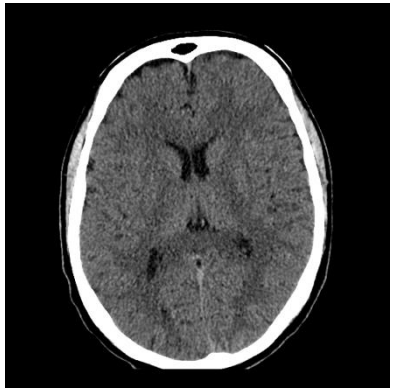
T1-weighted



T1 + gadolinium (tumour)



T2-weighted (coronal)



CT, for comparison

Nishimura, Principles of Magnetic Resonance Imaging (textbook); Brown et al., MRI: Physical Principles and Sequence Design (textbook).

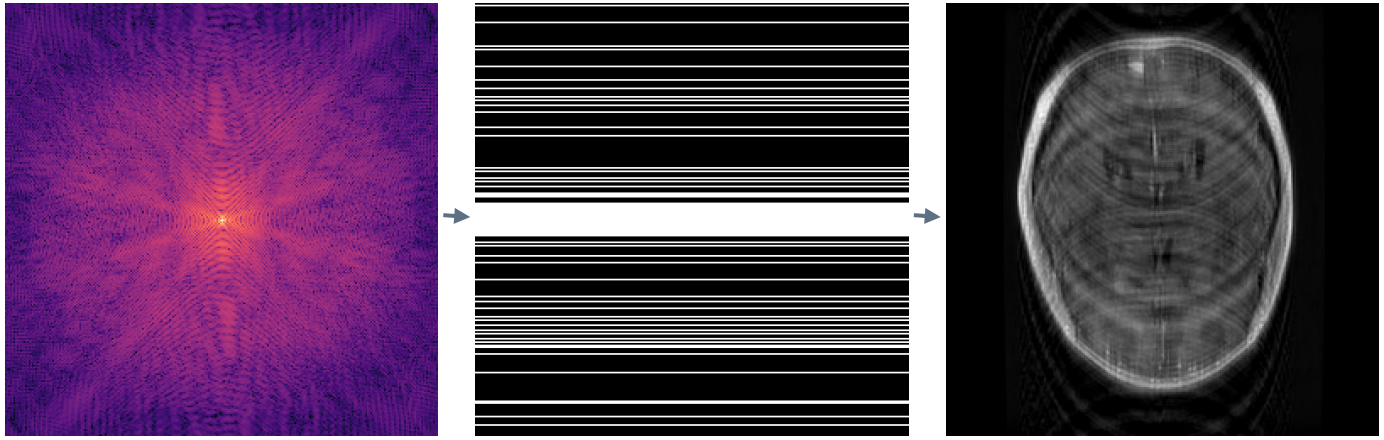
k-Space and the MRI Forward Model

MRI measures Fourier coefficients; skipping them makes reconstruction an ill-posed inverse problem

$$y = P\,F\,S\,x + \varepsilon$$

x	target image (complex), N pixels
S	coil sensitivity operator (one map per coil)
F	discrete Fourier transform
P	sampling operator (keeps acquired k-space points)
y	measured k-space samples
ε	complex Gaussian thermal noise

- **Nyquist:** full grid ⇒ $x = F^{-1}y$ (single coil)
- **Uniform R× skipping:** coherent fold-over aliasing
- **Random skipping:** incoherent, noise-like aliasing
- **Ill-posed:** P removes information ⇒ non-unique x



log |k-space| | mask P, R = 4 | $F^{-1}(Py)$: aliasing

Scan time ∝ number of phase-encode lines. R = 4 cuts acquisition time roughly fourfold — and removes 75% of the measurements the image depends on.

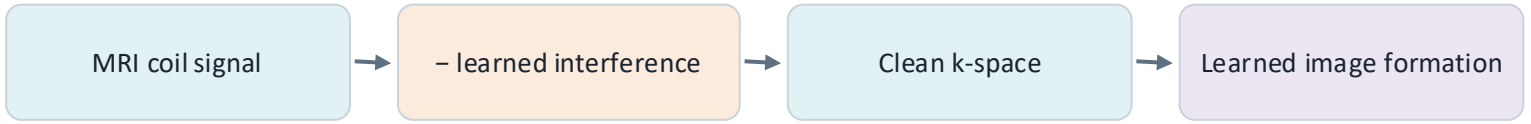
Simulated from a real T1 slice: single coil (S = I), σ = 0.002, 20 central + 44 random lines.

Pruessmann et al., SENSE, Magn. Reson. Med. 1999; Lustig, Donoho & Pauly, Sparse MRI, Magn. Reson. Med. 2007, doi:10.1002/mrm.21391.

Paper Analysis: Whole-Body MRI at 0.05 T

Science 2024 — AI changes the instrument itself, not only the analysis of its images

Element	Whole-body MRI at 0.05 T (Science 2024)
Scientific problem	MRI needs costly high-field magnets and shielded rooms
Innovation	Shielding-free ultra-low-field scanner; learned interference cancellation from sensing coils
AI role	Predict and subtract interference; deep-learning image formation trained on high-field data
Evidence type	Technical feasibility; human imaging demonstrations
Limitations	Low SNR, long scans; learned priors may impose high-field-like detail
Contribution	AI as part of the physical instrument — access, not only accuracy



Link to Case Study 1

- A prior trained on high-field images can make low-field images look high-field
- Validate with task-based tests, not appearance

Discuss

- What would you measure before using it for stroke triage?

Zhao Y. et al., Whole-body magnetic resonance imaging at 0.05 Tesla, Science 2024. Summarised at the level of the main idea; verify specifics in the paper.

CT: Attenuation, Projections, and Dose

Log-transformed X-ray intensities are line integrals — the Radon transform of μ

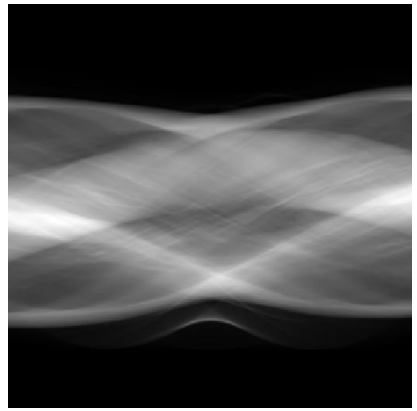
$$I = I_0 \exp\left(-\int_L \mu(s)\, ds \right) \quad \Rightarrow \quad p = -\ln(I/I_0) = \int_L \mu(s)\, ds$$

I_0	incident photon intensity (∝ dose)
I	detected intensity
$\mu(s)$	linear attenuation along ray L
p	projection value (sinogram entry)

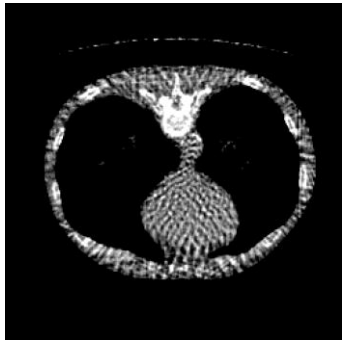
- **Radon transform:** p(θ, t) over angles θ and detector offsets t
- **Filtered backprojection (FBP):** ramp filter + backprojection (Fourier slice theorem)
- **Iterative reconstruction:** statistical noise model + regulariser
- **Dose–noise trade-off:** Var(p) ≈ 1/I, image σ ∝ 1/√dose



Real chest CT (reference)



Simulated sinogram



45 views: streaks



Low dose: noise

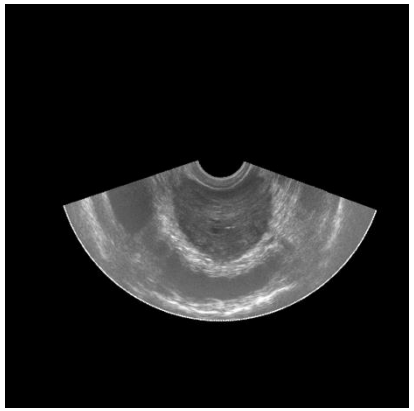
Kak & Slaney, Principles of Computerized Tomographic Imaging (IEEE Press). Real chest CT (3D Slicer sample data); acquisitions simulated. PSNR full 29.49 dB, 45 views 21.52 dB, low dose 24.42 dB.

Ultrasound and PET

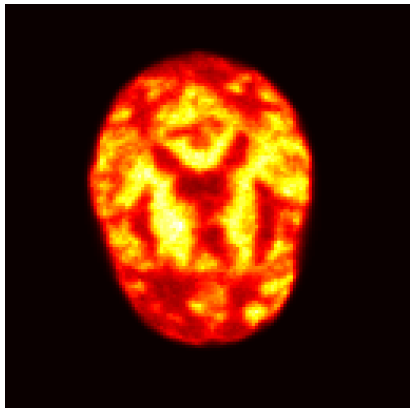
Different physics, different noise: multiplicative speckle versus Poisson photon counting

Ultrasound

- Pulse–echo; depth $z = c \cdot t/2$ ($c \approx 1540$ m/s)
- Reflection $R = ((Z_2 - Z_1)/(Z_2 + Z_1))^2$, $Z = \rho c$
- Delay-and-sum beamforming over array elements
- Speckle: Rayleigh envelope, multiplicative
- AI: learned beamforming, despeckling, guidance



Real B-mode (prostate)

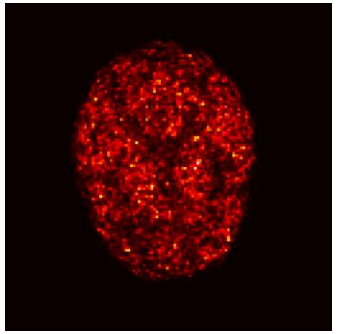


Simulated PET, 5×10^6 counts

PET

- β^+ annihilation → two 511 keV photons
- Coincidences define lines of response (LOR)
- $y \sim \text{Poisson}(Ax + r)$, r = randoms + scatter
- MLEM: $\lambda \leftarrow \lambda / (A^T 1) \cdot A^T[y / (A\lambda + r)]$
- Low counts → noise; AI: low-dose denoising

Loss functions should match noise physics: L2 assumes Gaussian noise; Poisson deviance for PET counts.



Simulated PET, 5×10^4 counts

Szabo, Diagnostic Ultrasound Imaging (textbook); Shepp & Vardi, IEEE Trans. Med. Imaging 1982 (ML-EM); Hudson & Larkin, IEEE TMI 1994 (OSEM).

Weighted Images versus Quantitative Maps

An imaging biomarker is a measurement: characterise bias, precision, and repeatability like an instrument

Weighted images

- T1w, T2w, FLAIR: arbitrary units
- Scanner-, coil-, protocol-dependent
- Excellent for detection and pattern
- Not directly comparable across sites

Quantitative maps

- T1, T2 (ms) · ADC (mm^2/s) · Ktrans (min^{-1})
- PET SUV, kinetic rates · CT HU
- Forward model with few physical parameters
- Comparable across time and sites if calibrated

Quantity	Unit	Forward model	AI role
T1, T2 relaxation	ms	Bloch / signal equations	Fast fitting; MR fingerprinting
ADC, microstructure	mm^2/s	Diffusion signal models	Parameter estimation, denoising
Ktrans (DCE-MRI)	min^{-1}	Tofts pharmacokinetic model	Robust fitting, AIF estimation
PET SUV, Ki	g/mL, mL/min/g	Compartment kinetics	Low-count estimation
CT attenuation	HU	Beer–Lambert	Low-dose recon preserving HU

measurement model

$$Y = \theta + b + e, \quad e \sim \mathcal{N}(0, \sigma_{w^2}), \quad RC = 2.77\ \sigma_w$$

θ	true value of the quantity
b	bias (needs a reference standard or phantom)
σ_w	within-subject SD from test–retest scans
RC	95% limit for test–retest differences

Worked number: σ_w = 30 ms ⇒ RC = 83 ms. A 60 ms change in T1 between visits is within measurement noise.

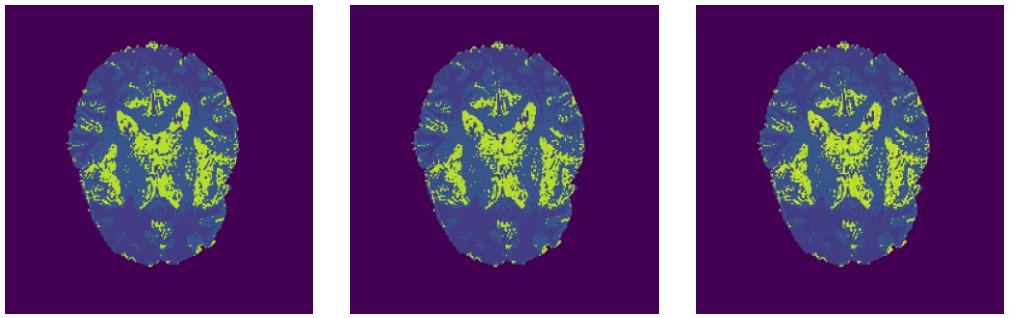
QIBA 'profiles' state a claim (e.g. 'a change > RC is real with 95% confidence') and the conditions under which it holds.

Sullivan et al., Metrology standards for quantitative imaging biomarkers, Radiology 2015; Raunig et al., Stat. Methods Med. Res. 2015; Ma et al., Magnetic resonance fingerprinting, Nature 2013.

Worked Example: Learned T1 Mapping

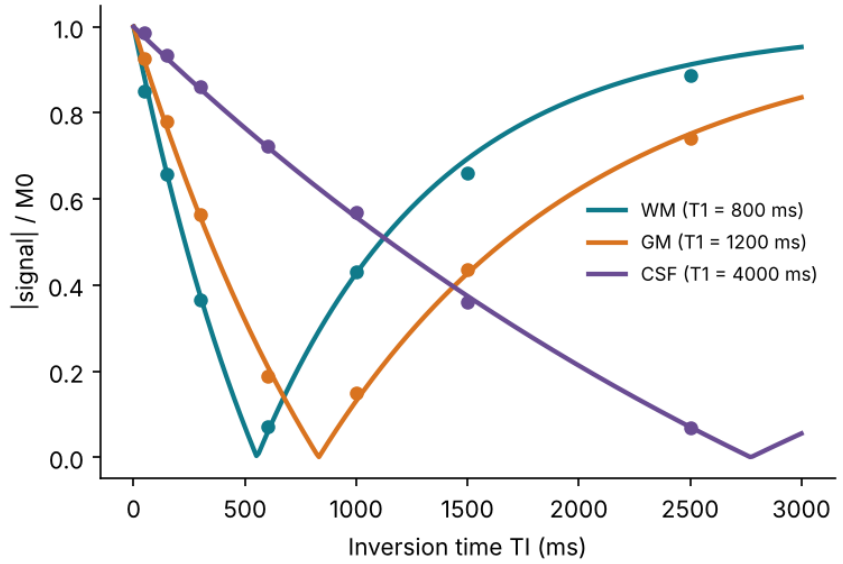
A neural estimator trained only on physics simulations — faster, but is it a better instrument?

$$S(TI) = | M_0 (1 - 2\, e^{-TI/T_1}) | + \text{noise}$$



True T1 map | Dictionary fit | Neural estimator

Tissue	T1 (ms)	Dict bias	Dict RC	NN bias	NN RC
WM	800	-0.1%	34 ms	+1.6%	62 ms
GM	1200	-0.1%	48 ms	-0.3%	126 ms
CSF	4000	+0.3%	261 ms	+1.5%	567 ms



Curves: noiseless model. Dots: one noisy voxel per tissue.

What the computation shows

- Neural estimator ≈ 6× faster than dictionary matching
- Both ≈ unbiased in WM/GM; NN repeatability ≈ 2× worse
- CSF worst for both: longest TI < T1 ⇒ design limits identifiability

Computed here: inversion recovery, 7 TIs, Rician noise σ = 0.02; tissue map by k-means on a real T1 slice, approximate T1 values. Cohen, Zhu & Rosen, MR fingerprinting deep reconstruction network (DRONE), Magn. Reson. Med. 2018.

Reconstruction as a Regularised Inverse Problem



Data consistency says what the measurements allow; the regulariser says what anatomy is plausible

$$x^* = \operatorname{argmin}_x \; \tfrac{1}{2}\|Ax - y\|_2^2 \; + \; \lambda \, R(x)$$

A	forward operator (e.g. PFS, Radon)
$\|Ax-y\|^2$	data consistency (Gaussian log-likelihood)
$R(x)$	regulariser = negative log-prior
λ	trade-off weight; chosen by validation

Tikhonov

$R = \|x\|^2$ — smooth, closed form, blurs edges

Total variation

$R = \|\nabla x\|_1$ — piecewise-constant, staircasing

Wavelet sparsity

$R = \|\Psi x\|_1$ — basis of compressed sensing

Learned prior

R or its proximal map learned from data

Why ill-posed?

- **Non-uniqueness:** null space of A — components invisible to y
- **Instability:** small singular values amplify noise
- **Bayesian view:** x* is a MAP estimate; R encodes prior belief
- **Example:** R = 4 Cartesian MRI keeps 64/256 lines ⇒ 192-dimensional null space per image column
- **Consequence:** the prior, not the data, fills in what was never measured

Lustig et al., Magn. Reson. Med. 2007; Candès, Romberg & Tao, IEEE Trans. Inf. Theory 2006 (compressed sensing).

Learning-Based Reconstruction: A Taxonomy



From direct mappings to physics-guided learning: how much of A should the network see?

Approach	Formulation	Uses A at inference?	Strength	Main risk
Direct inversion	$\hat{x} = f\theta(y)$	No	Fast, no physics model needed	Data hungry; fragile to protocol shift
Image-domain post-processing	$\hat{x} = f\theta(A^H y)$	Only for initial image	Simple; reuse U-Net	Cannot enforce data consistency
Unrolled optimisation	K learned iterations with $A^H(Ax-y)$	Yes, every iteration	Strong accuracy per parameter	Fixed K; memory grows with K
Plug-and-play prior	proximal step := pretrained denoiser	Yes	Modular; no end-to-end retraining	Convergence needs denoiser conditions
Deep equilibrium	$x = T\theta(x; y)$ fixed point	Yes	Constant memory; 'infinite' depth	Fixed-point solve can be slow
Diffusion inverse solver	sample $p(x \mid y)$ with learned score + DC	Yes	Posterior samples; uncertainty	Plausible hallucinations; slow sampling

Trend: keep the physics explicit. Measured data should constrain what the network is allowed to invent.

Zhu et al., AUTOMAP, Nature 2018; Aggarwal et al., MoDL, IEEE TMI 2019; Venkatakrishnan et al., Plug-and-Play priors, GlobalSIP 2013; Gilton et al., Deep equilibrium, IEEE TCI 2021; Chung & Ye, Med. Image Anal. 2022.

One Mathematics, Many Sciences

The same inverse-problem structure appears wherever science measures indirectly

$$y = A(x) + \varepsilon \quad \rightarrow \quad p(x \mid y) \propto p(y \mid x) \cdot p(x)$$

Domain	Unknown x	Forward operator A	Noise	AI / prior	Ground truth?
MRI	Magnetisation image	Undersampled Fourier (PFS)	Complex Gaussian	Unrolled nets, diffusion priors	Fully sampled scans
CT / PET	Attenuation / tracer density	Radon / system matrix	Poisson	Learned iterative, MLEM + priors	Phantoms, high-dose scans
Radio interferometry	Sky brightness (black hole)	Sparse Fourier samples	Gaussian + calibration	Regularised & generative priors	None: synthetic tests, method agreement
Cryo-EM	3D molecular density	Projections, unknown poses	Very low SNR	Bayesian refinement, learned priors	Half-map consistency (FSC)
Seismology	Subsurface velocity	Wave equation	Gaussian	Full-waveform inversion, neural operators	Boreholes, synthetic models
Microscopy	Fluorophore density	PSF blur + sampling	Poisson–Gaussian	Learned restoration	Paired high-SNR acquisitions

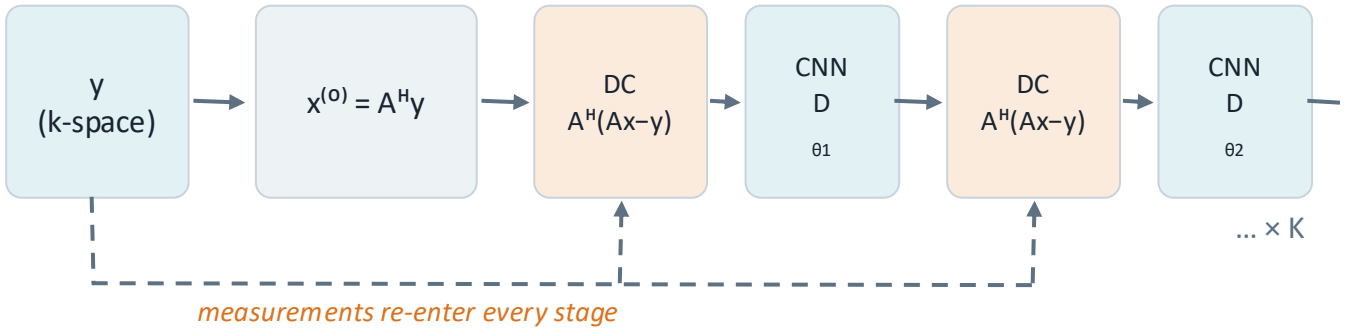
Shared lessons: priors fill the null space · report posterior uncertainty · keep the physics explicit · without ground truth, validate by simulation, held-out data, and agreement between independent methods.

Event Horizon Telescope Collaboration, Astrophys. J. Lett. 2019 (M87 imaging); Scheres, RELION, J. Struct. Biol. 2012; Weigert et al., Content-aware image restoration, Nat. Methods 2018; Li et al., Fourier neural operator, ICLR 2021.

Unrolled Reconstruction Networks

Turn K iterations of a solver into K network stages with learned denoisers and data-consistency blocks

$$x^{(k+1)} = D_{\theta k}\left(x^{(k)} - \eta\, A^{H}(A\, x^{(k)} - y) \right)$$



- **Learned denoiser** $D_{\theta k}$ replaces the proximal map of λR
- **Data consistency** keeps the estimate faithful to y
- **Training:** $L = \|\hat{x} - x_{ref}\|_1 + \alpha(1 - SSIM)$
- **Plug-and-play:** pretrained D, no unrolled training
- **Deep equilibrium:** iterate one stage to a fixed point

```
# Unrolled reconstruction (inference)
input: y, mask P, coil maps S, K stages
x ← S^H F^H P^T y          # zero-filled init
for k in 1..K:
   r  ← P F S x − y        # k-space residual
   g  ← S^H F^H P^T r      # gradient of DC
   z  ← x − η_k · g
   x  ← z + CNN_θk(z)      # residual denoiser
return |x|

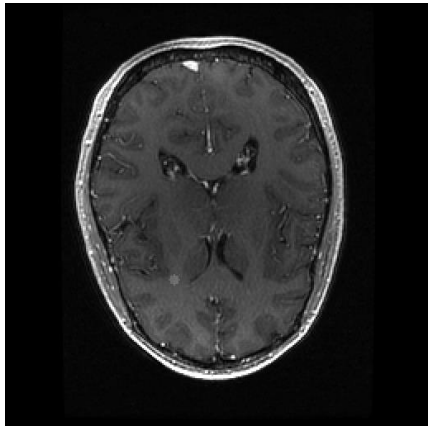
# Training
for (y, x_ref) in loader:
   loss ← L1(x^, x_ref) + α(1−SSIM)
   update θ, η by backprop through all K
```

Hammernik et al., Variational network, Magn. Reson. Med. 2018, doi:10.1002/mrm.26977; Sriram et al., End-to-end variational networks, MICCAI 2020.

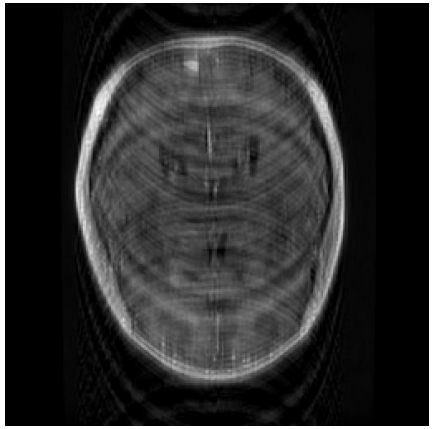
Case Study 1: Accelerated MRI (R = 4)

Reconstruct

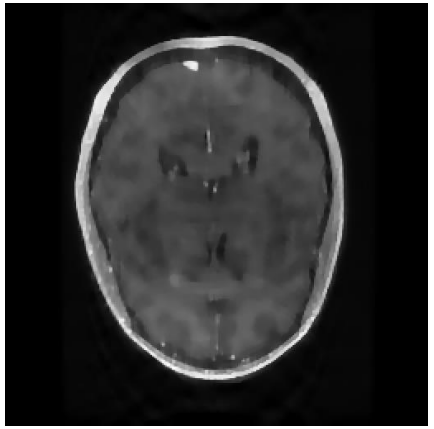
≈ 2 min



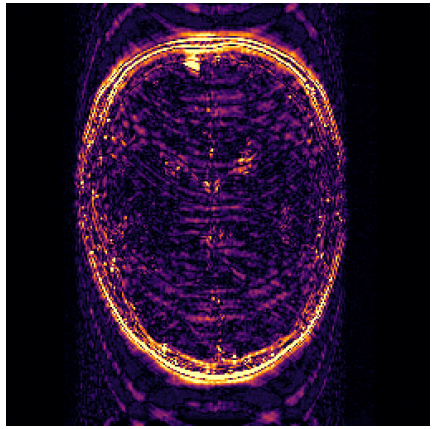
Fully sampled reference



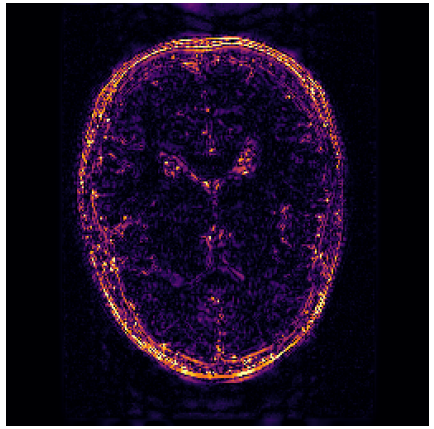
Zero-filled $F^{-1}(Py)$



TV-regularised CS



|error| zero-filled



|error| TV-CS

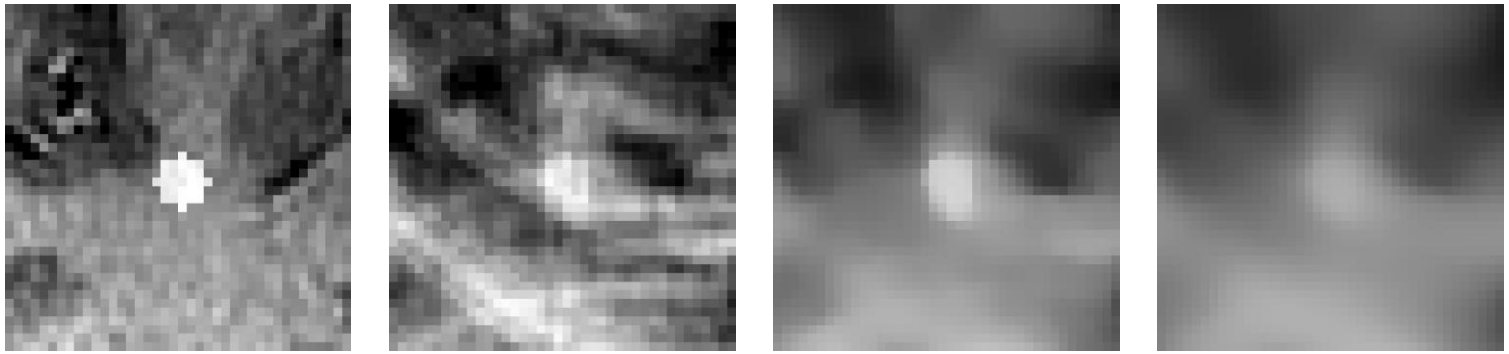
Reconstruction	MSE	PSNR (dB)	SSIM	Status
Zero-filled inverse FFT	0.005	23.46	0.717	computed
TV-regularised CS (λ = 0.03)	1.75e-3	27.57	0.857	computed
Learned unrolled network	—	—	—	not trained here; see fastMRI literature

MSE = $(1/N)\ \Sigma\ (\hat{x}_i - x_i)^2$
PSNR = $10\ \log_{10}(MAX^2\ /\ MSE)$
SSIM = [l · c · s] local luminance, contrast, structure
Single coil, retrospective, MAX = 1

Metrics computed here (skimage, data range 1). Learned-reconstruction performance on real data: Zbontar et al., fastMRI, arXiv:1811.08839; Muckley et al., IEEE TMI 2021 (2020 fastMRI challenge).

Higher PSNR ≠ Preserved Lesion

Global fidelity metrics average over pixels; a 3-pixel lesion barely moves them



Reference | Zero-filled | TV-CS (λ = 0.03) | TV-CS (λ = 0.12)

Image	PSNR (dB)	Lesion contrast	% of true
Reference	∞	0.135	100%
Zero-filled	23.46	0.105	78%
TV-CS λ = 0.03	27.57	0.075	56%
TV-CS λ = 0.12	24.21	0.036	27%

Image quality ≠ diagnostic accuracy

- PSNR/SSIM average over 65,536 pixels; the lesion is 29
- Strong prior raises PSNR yet erases the finding
- Learned priors can also hallucinate structure

Task-based evaluation

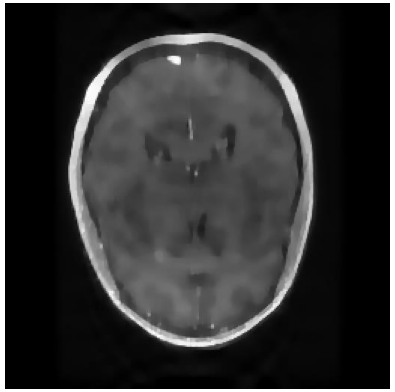
- Lesion detectability (model or human observer)
- Downstream segmentation / measurement error
- Blinded reader study on confirmed lesions

Computed here. Antun et al., On instabilities of deep learning in image reconstruction, PNAS 2020, doi:10.1073/pnas.1907377117.

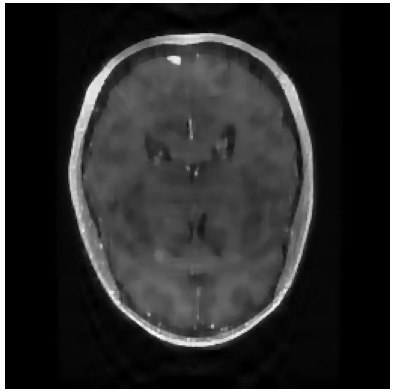
From One Image to a Distribution of Images

Reconstruction from incomplete data is inference: report the spread, not only the best image

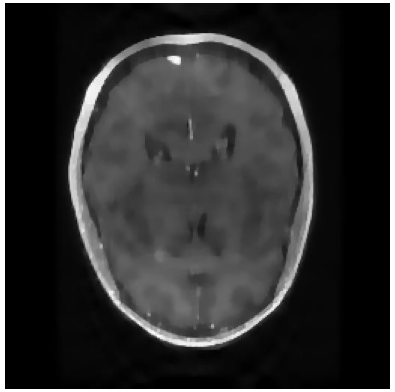
$$p(x \mid y) \propto p(y \mid x) \cdot p(x) \quad \text{MAP} = \operatorname{argmax}_x p(x \mid y) \quad \text{uncertainty} = \text{spread of samples}$$



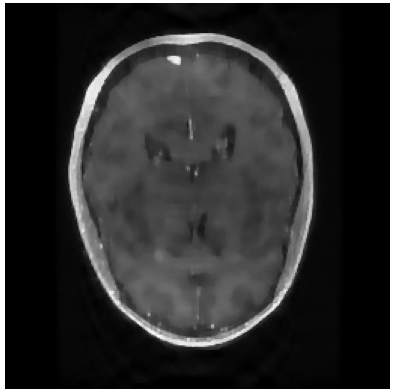
Sample 1



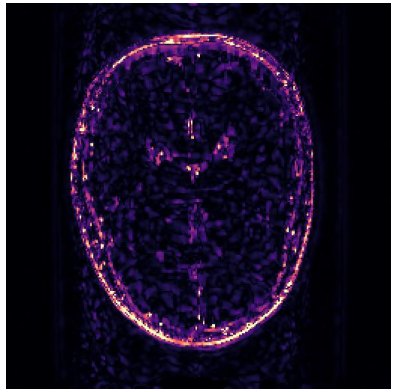
Sample 2



Sample 3



Mean of 8



Pixelwise SD

Quantity (computed)	Value
SD inside lesion / surrounding ring	0.0095 / 0.0023 (≈ 4×)
Lesion contrast across samples (truth 0.135)	0.059 – 0.089

What this is: a randomised-MAP ensemble (noise, λ, initialisation perturbed) — a proxy, not a calibrated posterior. Diffusion posterior sampling does this properly; coverage must be checked on simulations.

Computed here on Case Study 1 data. Papandreou & Yuille, Perturb-and-MAP, ICCV 2011; Song et al., Solving inverse problems in medical imaging with score-based generative models, ICLR 2022; Chung & Ye, Med. Image Anal. 2022.

Convolutional Neural Networks

Local, weight-shared filters give translation equivariance and parameter efficiency

$$z_{c'}(i, j) = b_{c'} + \Sigma_c \Sigma_{u,v} w_{c',c}(u, v) \cdot x_c(i + u, j + v)$$

x_c	input feature map, channel c
$w_{c',c}$	k × k kernel from channel c to c′
$b_{c'}$	bias of output channel c′
$z_{c'}$	pre-activation output; then ReLU, norm

- **Parameters:** 3×3 conv, 64→128 channels: 3·3·64·128 + 128 = 73,856 (independent of image size)
- **Equivariance:** shift(x) ⇒ shift(z); pooling gives local invariance
- **Stride / pooling** reduce resolution and enlarge the receptive field
- **Inductive bias:** locality + weight sharing ⇒ data-efficient on small medical datasets

Worked example: edge filter

x_1	x_2	x_3	x_4
0	0	1	1
0	0	1	1
0	0	1	1

*

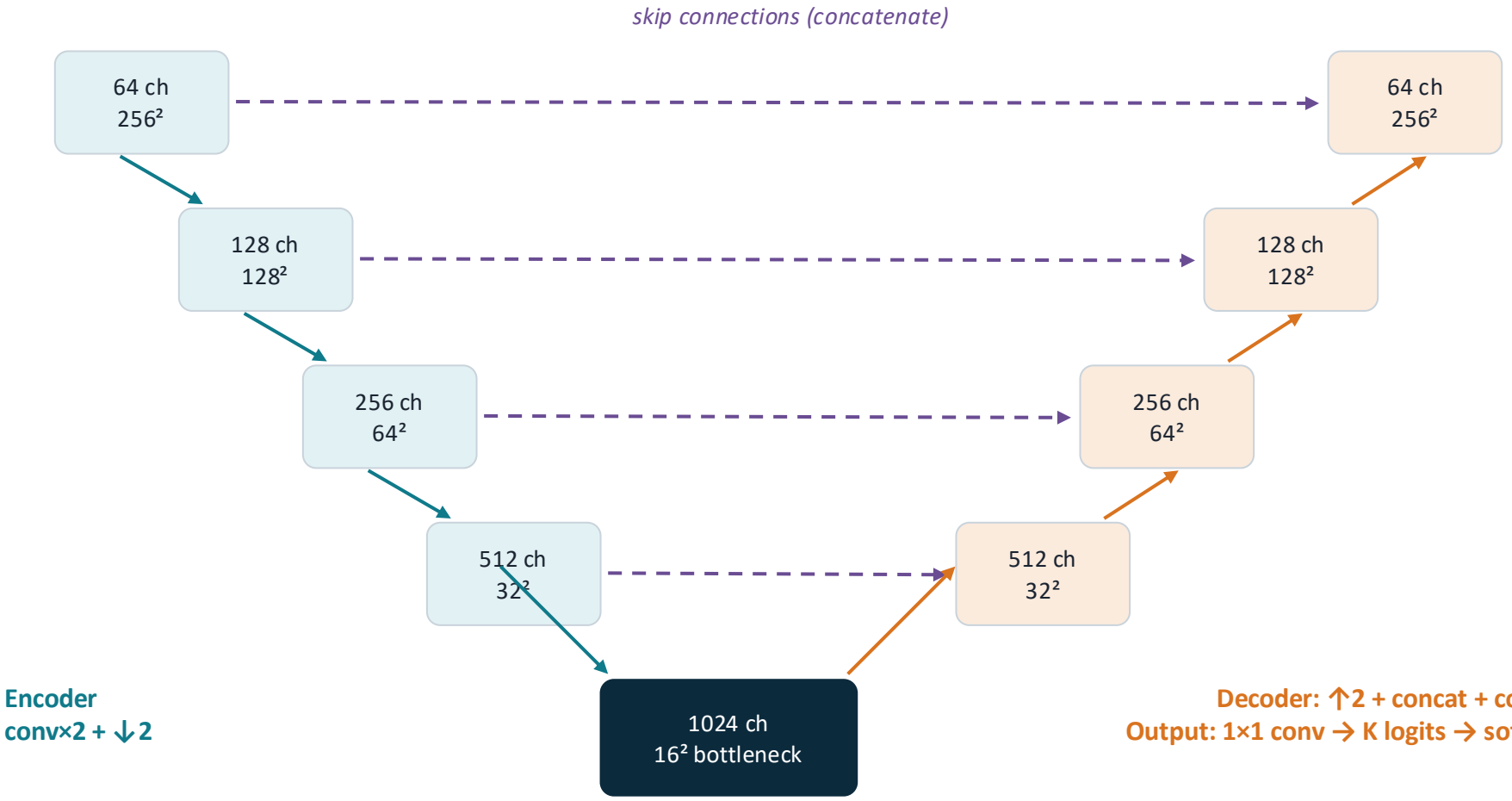
k_1	k_2	k_3
−1	0	1
−1	0	1
−1	0	1

Two valid outputs (centred at columns 2 and 3):
z = (−0−0−0 + 1+1+1) = 3, z = (−0−0−0 + 1+1+1) = 3
On a uniform patch: z = 0.
The kernel responds only where intensity changes horizontally — a learned version of this is a first-layer feature.

LeCun et al., Proc. IEEE 1998; Goodfellow, Bengio & Courville, Deep Learning, MIT Press 2016, ch. 9.

U-Net: Encoder, Decoder, Skip Connections

Context from the encoder, localisation from skip connections — pixelwise prediction at full resolution



Why it works

- Multi-scale context in the encoder
- Skips restore boundary detail
- Fully convolutional: any input size
- Strong with few hundred labelled cases

Practical choices

- Instance norm, leaky ReLU
- Deep supervision at decoder levels
- 3D patches (e.g. 128^3) for volumes
- Loss: Dice + cross-entropy

Ronneberger, Fischer & Brox, U-Net, MICCAI 2015, doi:10.1007/978-3-319-24574-4_28; Çiçek et al., 3D U-Net, MICCAI 2016.

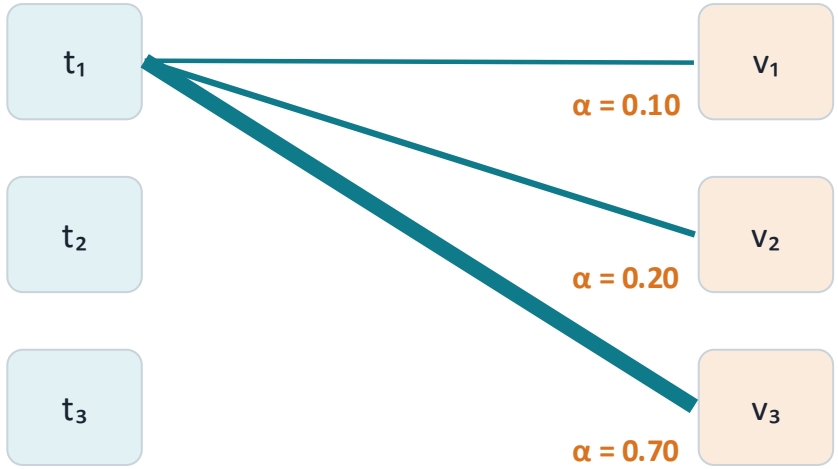
Self-Attention and Transformers

Every token can attend to every other token — global context with content-dependent weights

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$$

$Q = XW_Q$	queries: what each token looks for
$K = XW_K$	keys: what each token offers
$V = XW_V$	values: content passed on
$\sqrt{d_k}$	keeps logit variance ≈ 1; avoids saturated softmax
h heads	parallel subspaces, concatenated, projected

- **Patch embedding:** 16×16 patches → linear projection to d
- **Positional encoding:** learned or sinusoidal; location matters in anatomy
- **Block:** LayerNorm → MHSA → residual → MLP → residual



$$o_1 = 0.10\, v_1 + 0.20\, v_2 + 0.70\, v_3$$

Row 1 of softmax($QK^T/\sqrt{d_k}$): token 1 attends mostly to token 3 (e.g. a contralateral region used for symmetry comparison).

Vaswani et al., Attention is all you need, NeurIPS 2017; Dosovitskiy et al., An image is worth 16×16 words (ViT), ICLR 2021.

Inductive Bias versus Data

Segment · Detect · Diagnose

Strong priors win with little data; flexible models win only with enough data or pretraining

	CNN	U-Net	Vision Transformer	Swin Transformer	Hybrid CNN–Transformer
Inductive bias	Locality, equivariance	+ multi-scale, skips	Minimal	Local windows, hierarchy	Local + global
Context	Grows with depth	Large at bottleneck	Global from layer 1	Grows via shifts	Global at chosen stages
Data need (from scratch)	Low–moderate	Low–moderate	High	Moderate–high	Moderate
Cost scaling	Linear in pixels	Linear	Quadratic in tokens	Linear in tokens	Mixed
Dense prediction	Needs decoder	Native	Needs decoder	With decoder	Native (UNETR-style)
Typical medical role	Classification	Segmentation baseline	Pretrained encoders	3D segmentation	3D segmentation

Rule of thumb: with hundreds of labelled volumes, a well-configured U-Net is the baseline to beat. Report it.

Isensee et al., nnU-Net revisited: a call for rigorous validation in 3D medical image segmentation, MICCAI 2024 — reports strong CNN baselines matching or exceeding many Transformer variants.

Classification and Detection

Image-level labels answer ‘is something wrong?’; detection answers ‘where?’

Classification: abnormal brain MRI

- Input: T1/T2/FLAIR volume; output p(abnormal)
- Loss: $L = -[y \log p + (1-y) \log(1-p)]$
- Metrics: sensitivity, specificity, AUROC
- Triage: sensitivity at fixed high specificity
- Risk: shortcut learning from scanner or text markers

Detection: pulmonary nodules in CT

- Output: 3D boxes (centre, diameter, score)
- Match by centre distance / IoU to reference
- Metric: FROC — sensitivity vs false positives per scan
- LUNA16 CPM: mean sensitivity at 1/8…8 FP/scan
- Imbalance: focal loss, hard-negative mining

focal loss

$$\mathrm{FL}(p_t) = -(1 - p_t)^{\gamma} \log p_t$$

p_t: model probability of the true class; $\gamma \geq 0$ down-weights easy negatives. With $\gamma = 2$, an easy background voxel with $p_t = 0.9$ is weighted $(0.1)^2 = 0.01$.

Unit of analysis changes the answer: image-, lesion-, and patient-level metrics are different experiments.

Setio et al., Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in CT (LUNA16), Med. Image Anal. 2017; Lin et al., Focal loss, ICCV 2017.

Paper Analysis: PANDA (Nature Medicine 2023)



Detecting a cancer on scans where clinicians usually cannot see it

Element	PANDA (Nature Medicine 2023)
Scientific problem	Pancreatic cancer found late; usually invisible on non-contrast CT
Innovation	Deep learning detects lesions on routine non-contrast CT
Labels	Informed by contrast-enhanced imaging of the same patients (verify)
Evidence type	Retrospective multicentre + large real-world consecutive study
Limitations	Not a prospective screening trial; population and scanner scope
Contribution	Signal present but imperceptible to readers — opportunistic screening

$$PPV = Se \cdot p \,/\, (Se \cdot p + (1 - Sp)(1 - p))$$

Screening arithmetic

- p = 1/1,000, Se = 0.90, Sp = 0.99
- PPV ≈ 8%: most positives are false
- Prevalence dominates screening value

AI-for-Science angle

- Information exists that experts cannot perceive
- Claims ladder: rungs 1–2, not yet 5

Cao K. et al., Large-scale pancreatic cancer detection via non-contrast CT and deep learning, Nat. Med. 2023. Numbers as reported; verify in the paper.

Segmentation and Its Metrics

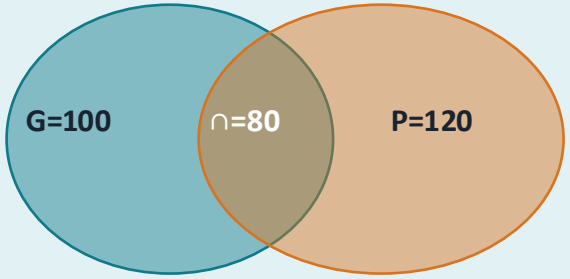


Overlap metrics reward volume agreement; distance metrics catch boundary and outlier errors

$$\text{Dice} = 2|P \cap G| \,/\, (|P| + |G|) \qquad \text{IoU} = |P \cap G| \,/\, |P \cup G| = \text{Dice} \,/\, (2 - \text{Dice})$$

Metric	Definition	Sensitive to	Caveat
Dice / IoU	Overlap ratio	Volume agreement	Unstable for tiny structures
Hausdorff (HD95)	95th pct. max surface distance	Outliers, boundaries	Undefined if one set empty
ASSD	Mean symmetric surface distance	Average boundary error	Hides rare large errors
Sensitivity	TP / (TP + FN)	Missed tissue	Voxel vs lesion level differ
Specificity	TN / (TN + FP)	False positives	≈ 1 at voxel level (background)
AUROC	Area under ROC over thresholds	Ranking quality	Ignores calibration and prevalence

Worked example



Dice = 2·80 / 220 = 0.727
IoU = 80 / 140 = 0.571
Sensitivity = 80/100 = 0.80
Precision = 80/120 = 0.667

Small-lesion check: a 10-voxel lesion with 5 voxels found ⇒ Dice ≈ 0.67.

Maier-Hein et al., Metrics reloaded: recommendations for image analysis validation, Nature Methods 2024, doi:10.1038/s41592-023-02151-z.

Image Registration

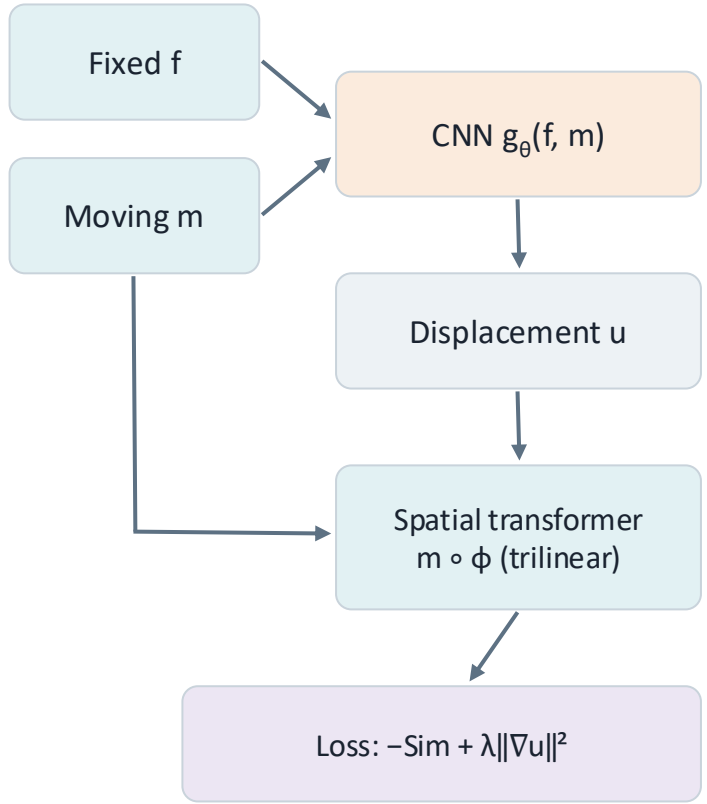
Estimate a spatial transform φ so that moving image m ∘φ aligns with fixed image f

$$\varphi^* = \arg\min_{\varphi} \; -\mathrm{Sim}(\, f, m \circ \varphi \,) \; + \; \lambda \, \|\nabla u\|^2, \quad \varphi(x) = x + u(x)$$

Transform	DOF (3D)	Use
Rigid	6	Intra-subject, multi-sequence
Affine	12	Inter-subject, global scaling
Deformable	3 per voxel	Atlas, longitudinal, soft tissue

Similarity	When
MSE	Same modality, same contrast
NCC	Linear intensity relation
Mutual information	Cross-modality (T1 ↔ FLAIR, CT ↔ MR)

Check folding: det $J_\phi(x) \le 0$ means the transform is not invertible there. Pathology (tumour, resection) violates smoothness assumptions.



Unsupervised: no ground-truth deformations needed

Balakrishnan et al., VoxelMorph, IEEE TMI 2019, doi:10.1109/TMI.2019.2897538; Jaderberg et al., Spatial transformer networks, NeurIPS 2015.

Radiomics and Deep Representations

Hand-crafted quantitative features are interpretable but fragile to acquisition and segmentation

Image → Segmentation → Feature extraction → Selection → Model → Validation

Shape

- Volume, surface area
- Sphericity = $\pi^{1/3}(6V)^{2/3} / A$
- Elongation, compactness

First-order

- Mean, variance, skewness
- Histogram entropy
- Percentiles

Texture

- GLCM contrast, homogeneity
- Run-length, size-zone
- Depends on discretisation

Robustness threats

- Voxel size, kernel, scanner
- Segmentation variability
- Hundreds of features, small n

Radiomics vs deep features

Radiomics: designed, standardised (IBSI), interpretable, low capacity.

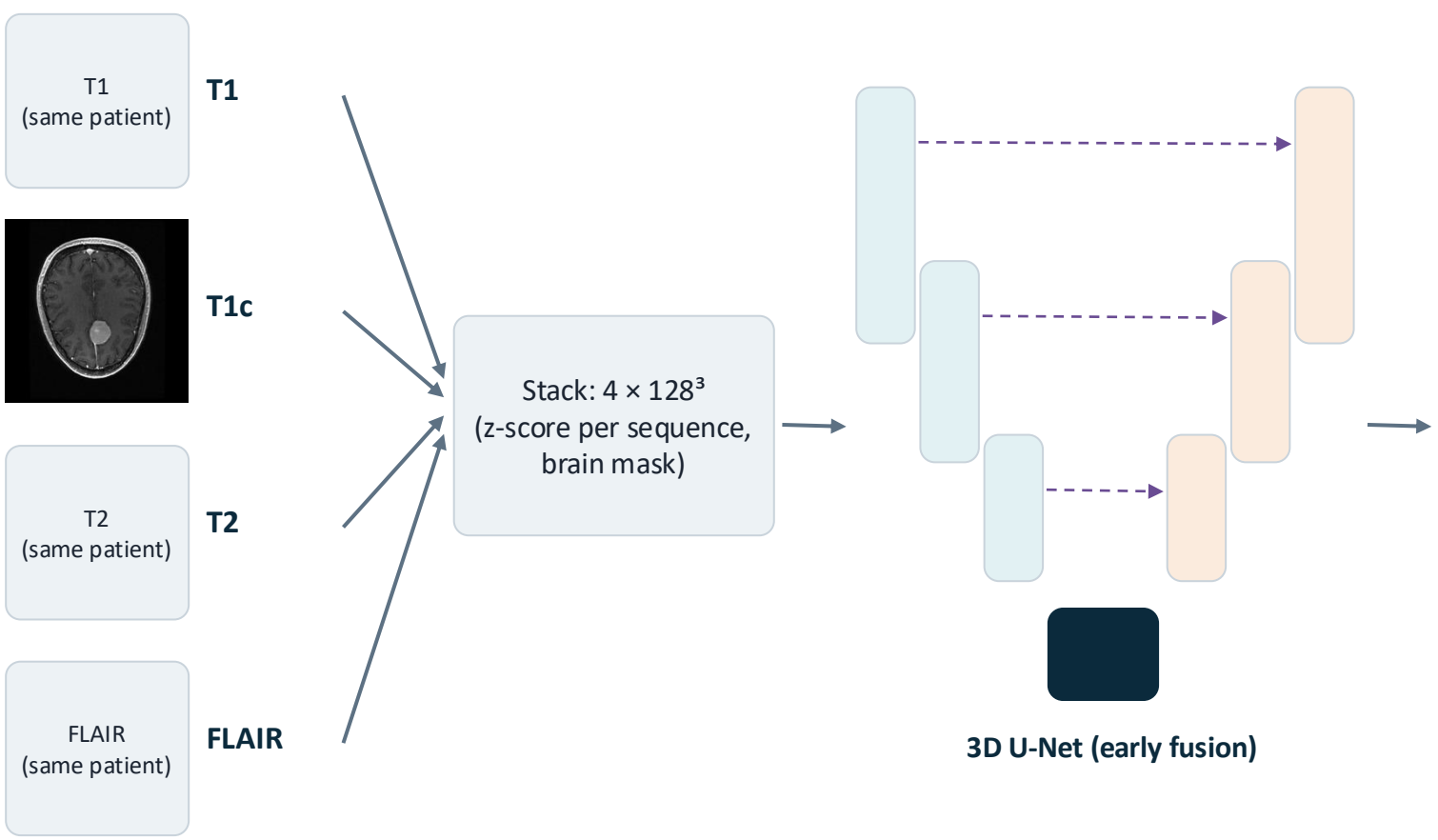
Deep: learned, high capacity, less interpretable.

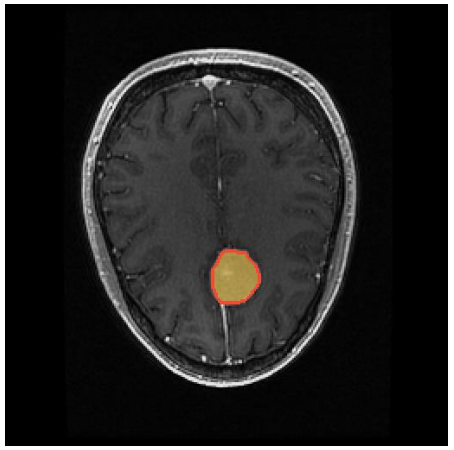
Both: sensitive to acquisition shift; both need external validation.

Lambin et al., Eur. J. Cancer 2012; Aerts et al., Nat. Commun. 2014, doi:10.1038/ncomms5006; Zwanenburg et al., IBSI, Radiology 2020, doi:10.1148/radiol.2020191145.

Case Study 2: Brain Tumour Segmentation

Four co-registered sequences in, three nested tumour sub-regions out





Real T1c, computed mask

Real enhancing tumour (3D Slicer sample). Mask: 3D region growing, not a trained model. 13,271 voxels = 16.3 cm³.

Evaluated region	Labels
Whole tumour (WT)	NCR + ET + ED
Tumour core (TC)	NCR + ET
Enhancing (ET)	ET

Menze et al., BraTS benchmark, IEEE TMI 2015, doi:10.1109/TMI.2014.2377694; Isensee et al., Nat. Methods 2021. Image: 3D Slicer sample data (MRBrainTumor1).

Case Study 2: Loss, Volume, and Validation Choices

Volumes need physical voxel spacing; validation needs patient-level splits

$$L = L_{Dice} + L_{CE}, \quad L_{Dice} = 1 - 2\Sigma_i p_i g_i / (\Sigma_i p_i + \Sigma_i g_i + \varepsilon)$$

$$V = N_{vox} \cdot \Delta x \cdot \Delta y \cdot \Delta z$$

Real volume computed for this deck.

Brief: 25,000 vox × 1 mm^3 = 25.0 cm^3; at 0.9×0.9×5 mm = 101 cm^3

Real tumour: 13,271 vox × 0.9375×0.9375×1.4 mm^3 = 16.3 cm^3 (threshold ±20%: 15.5–17.1)

Class imbalance

Tumour often < 1% of voxels: Dice term, patch oversampling of foreground

Normalisation

MRI intensities are not quantitative: z-score per sequence inside the brain mask

3D vs 2D

3D: through-plane context, more memory. Anisotropic data: 2.5D or anisotropic kernels

Missing data

Missing sequences: modality dropout in training; late fusion; synthesis (Module II) with caution

Volume as a measurement

- Here: the threshold choice alone moves the volume 15.5–17.1 cm^3
- Budget: segmentation, spacing, partial volume, scanner
- A change < RC between visits is not evidence of growth

Patient-level validation

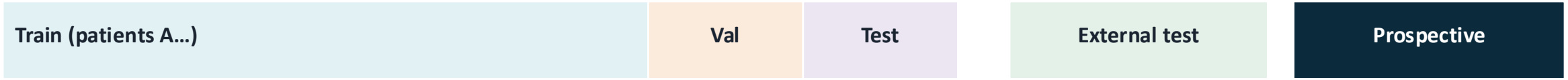
- Slice-level splits leak the test patient into training
- Metrics per patient, aggregated with CIs

Milletari et al., V-Net (soft Dice), 3DV 2016; Bakas et al., arXiv:1811.02629 (BraTS evaluation and sub-regions).

Evaluation Design: Leakage, Shift, Statistics

Cross-cutting

The test set must represent patients the model has never seen, from settings it will be used in



Leakage

- Same patient in train and test
- Tuning on the test set
- Normalisation stats from all data
- Labels from text the model sees

Shortcuts

- Scanner, site, markers predict label
- Prevalence differs across sites
- Zech et al. 2018: hospital system recognised

Shift & label noise

- Covariate, label, concept shift
- Inter-rater variability bounds accuracy
- Report per-site performance

Statistics

- Patient-level bootstrap 95% CIs
- Paired tests; seeds and folds
- Pre-specify the primary metric
- Calibration: see Module III

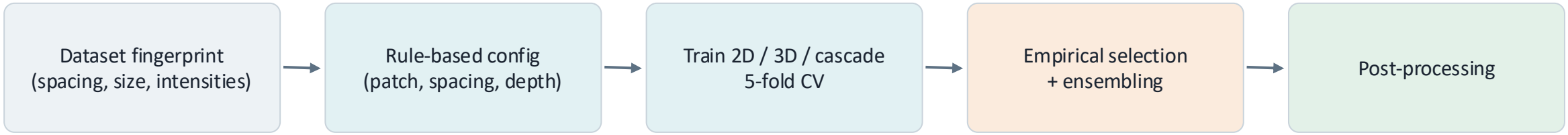
Same Dice, different claims: 0.86 [95% CI 0.82–0.89] on 60 external patients vs 0.86 on 6 internal patients. Internal → in-distribution · external → generalisation · reader study → utility · prospective → deployment.

Tejani et al., CLAIM 2024 Update, Radiol. Artif. Intell. 2024, doi:10.1148/ryai.240300; Zech et al., PLoS Med. 2018, doi:10.1371/journal.pmed.1002683.

nnU-Net: A Self-Configuring Baseline



Most performance comes from systematic configuration, not novel architecture



Analysis	nnU-Net (Isensee et al., Nature Methods 2021)
Problem	Manual, dataset-specific pipeline design limits reproducibility
Innovation	Fixed + rule-based + empirical parameters from a dataset fingerprint
Architecture / objective	Plain 2D/3D U-Nets; Dice + cross-entropy; deep supervision
Experimental design	Many public segmentation challenges; no manual tuning per task
Evidence type	Retrospective benchmarking (not clinical validation)
Limitations	Compute cost; fully supervised; no calibrated uncertainty
Contribution	Reference baseline; reframed what counts as an advance

Isensee et al., nnU-Net, Nature Methods 2021, doi:10.1038/s41592-020-01008-z; Isensee et al., nnU-Net revisited, MICCAI 2024.

Module I Synthesis

Acquire · Reconstruct · Segment

From physical measurement to quantitative information

AI transforms reconstructed medical images into quantitative information, but the transition from quantitative measurements to clinical interpretation requires additional reasoning and evidence integration.

Physics

Forward models define what is measured and how noise behaves

Reconstruction

Priors fill the null space; metrics can hide lost lesions

Representation

CNN, U-Net, Transformers: inductive bias vs data

Measurement & validation

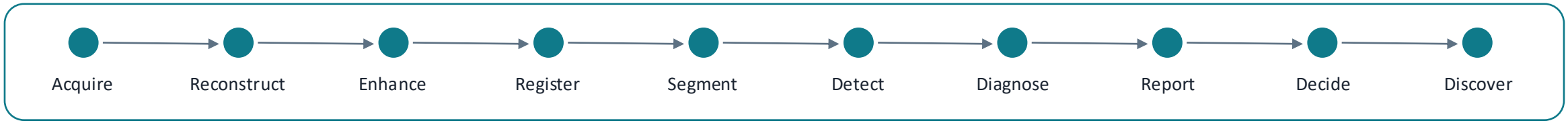
Bias, repeatability, uncertainty; patient-level splits, external tests

Conceptual questions

Q1. In Case Study 1, why can a reconstruction with higher PSNR be clinically worse? How would you test it?
Q2. A model reaches Dice 0.92 with a random 80/20 split of 2D slices. What is wrong, and how do you fix the protocol?

Module II — Foundation Models, Generative AI, and Multimodal Imaging

Self-supervision · foundation models · SAM / MedSAM / MedSAM2 · diffusion · vision–language models · report generation and its evaluation



Learning with Few Labels

Labels are the scarce resource; unlabelled images are abundant

Paradigm	Training signal	Medical example	Main risk
Supervised	Expert labels (x, y)	Voxel-wise tumour masks	Cost, inter-rater variability
Self-supervised	Pretext task on x only	Masked reconstruction of MRI patches	Pretext may ignore subtle pathology
Weak supervision	Noisy / coarse labels	Labels mined from radiology reports	Systematic label noise
Semi-supervised	Few labels + unlabelled	Pseudo-labels on unannotated scans	Confirmation bias
Transfer learning	Pretrained weights	ImageNet or medical pretraining	Domain gap (RGB vs 3D grey-scale)

Why labels are scarce

- Specialist time per 3D case
- Multiple raters to measure agreement
- Privacy limits pooling
- Rare diseases: few positives

What is abundant

- Archived unlabelled studies (PACS)
- Paired free-text reports
- Multiple sequences per patient

Azizi et al., Big self-supervised models advance medical image classification, ICCV 2021; Zhou et al., RETFound, Nature 2023, doi:10.1038/s41586-023-06555-x.

Contrastive Learning and InfoNCE

Cross-cutting

Pull two views of the same sample together; push other samples apart

$$\ell_{i,j} = -\log\left[\exp(\mathrm{sim}(z_i, z_j)/\tau) \,/\, \Sigma_{k\neq i} \exp(\mathrm{sim}(z_i, z_k)/\tau)\right]$$

z_i, z_j	embeddings of two augmented views of one image (positive pair)
z_k	embeddings of all other views in the batch or queue (negatives)
$\mathrm{sim}(u,v)$	cosine similarity $u^T v / (\|u\|\|v\|)$
τ	temperature; small τ emphasises hard negatives

Worked example	τ = 0.1	τ = 1.0
Logits (0.8, 0.3, 0.1)/τ	8, 3, 1	0.8, 0.3, 0.1
p(positive)	0.992	0.475
Loss ℓ	0.008	0.743

SimCLR

- Two augmentations, shared encoder
- Projection head; large batch of negatives

MoCo

- Momentum key encoder
- Queue of past keys = many negatives

Medical caveat

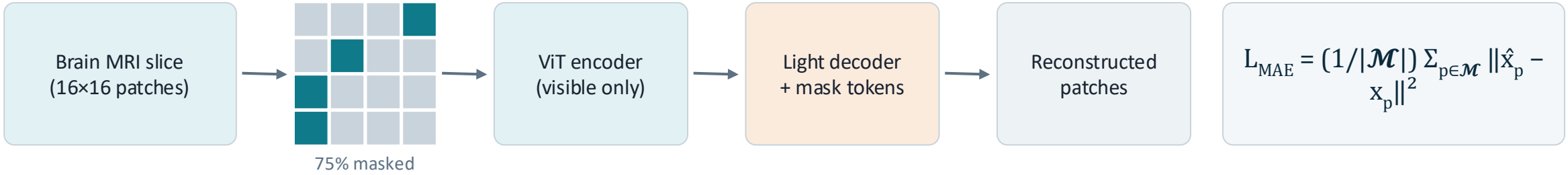
- Crops can remove the lesion
- Intensity jitter can mimic pathology

Oord et al., CPC, arXiv:1807.03748 (2018); Chen et al., SimCLR, ICML 2020; He et al., MoCo, CVPR 2020.

Masked Autoencoders and Image–Text Pretraining

Cross-cutting

Reconstruct what is hidden, or align images with the reports that describe them



Masked image modelling

- Mask 60–90% of patches (cubes in 3D)
- Loss only on masked patches
- Learns anatomy and texture structure
- Fine-tune encoder downstream

Image–text contrastive

- Image ↔ its report = positive pair
- Free supervision from clinical text
- Enables zero-shot labels via prompts
- Inherits report omissions and bias

Choosing a pretext

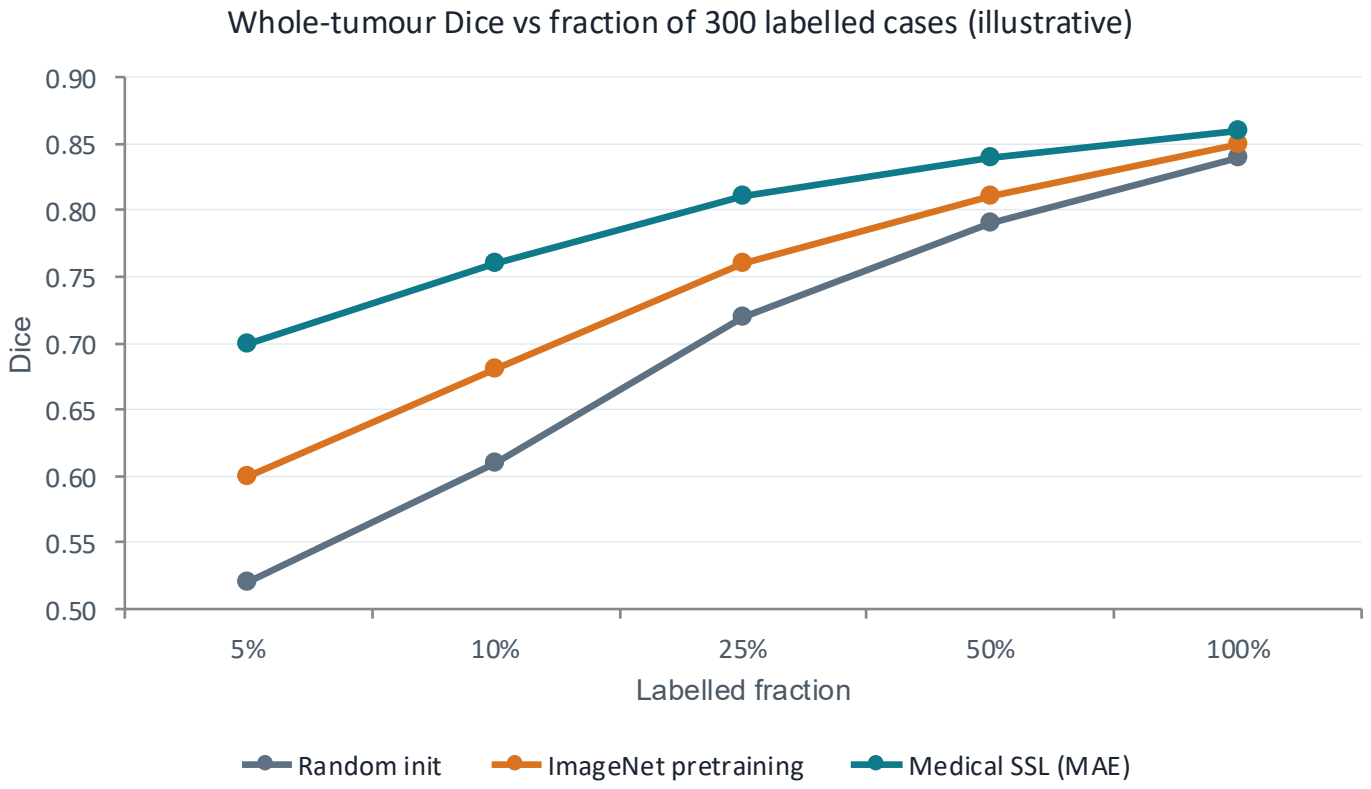
- Pixel-level tasks favour MAE-type
- Semantic retrieval favours image–text
- Hybrids combine both objectives
- Evaluate on the target clinical task

He et al., Masked autoencoders are scalable vision learners, CVPR 2022; Radford et al., CLIP, ICML 2021; Tiu et al., CheXzero, Nat. Biomed. Eng. 2022, doi:10.1038/s41551-022-00936-9.

Worked Example: Annotation Efficiency

Medical self-supervised pretraining matters most when labels are scarcest

ILLUSTRATIVE VALUES — not experimental results



Protocol

- Pretrain: masked 3D reconstruction, unlabelled MRI
- Fine-tune: 5–100% of 300 labelled cases
- Same budget, ≥3 seeds, patient-disjoint data

Reading the curve

- Largest gap at 5–10% labels
- SSL at 25% ≈ random init at 100% (illustrative)
- Gap narrows with abundant labels

Validity checks

- Pretraining corpus must exclude test patients
- Report error bars and external test

Paper Analysis: CT-FM (Nat. Mach. Intell. 2024)

A foundation model for cancer imaging biomarkers — self-supervised features judged as measurements

Element	CT-FM (Nature Machine Intelligence 2024)
Scientific problem	Task-specific biomarkers from small cohorts are fragile
Innovation	Self-supervised 3D contrastive pretraining on many CT lesions
Training data	Public lesion collection (DeepLesion), ~11k lesions as reported
Evaluation	Lesion site, malignancy, prognosis; label efficiency vs supervised and radiomics
Measurement checks	Feature stability and biological association analyses
Evidence type	Retrospective incl. external cohorts
Limitations	CT only; seed point needed; stability ≠ multi-scanner repeatability

Why it fits AI for Science

- Features evaluated like instruments
- Stability + biology, not only AUROC

Open question

- Is a learned feature a measurement before its bias and repeatability are known?

Pai S. et al., Foundation model for cancer imaging biomarkers, Nat. Mach. Intell. 2024. Scale as reported (~11,000 lesions from DeepLesion); verify specifics.

The Foundation Model Paradigm

Pretrain once at scale; adapt many times with little task-specific data



Zero-shot

No task labels; behaviour from pretraining and prompts (e.g. ‘no pleural effusion’)

Few-shot

A handful of labelled examples, via fine-tuning or in-context examples

Promptable

Output conditioned on points, boxes, or text supplied at inference

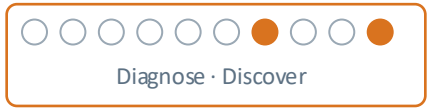
Parameter-efficient

Adapters, LoRA, prompt tuning: < 1–5% of weights trained

Medical requirements: native 3D and multi-sequence input · modality physics (HU in CT is quantitative; MRI intensity is not) · calibration · clinically defined tasks · external validation.

Bommasani et al., On the opportunities and risks of foundation models, arXiv:2108.07258 (2021); Moor et al., Nature 2023, doi:10.1038/s41586-023-05881-4.

Paper Analysis: RETFound (Nature 2023)



A retinal foundation model — and evidence that the eye carries systemic signals

Element	RETFound (Nature 2023)
Scientific problem	Many retinal and systemic tasks, few labels each
Innovation	Masked-autoencoder pretraining on ~1.6M unlabelled retinal images
Downstream	Ocular disease diagnosis/prognosis; systemic disease prediction (‘oculomics’)
Evidence type	Retrospective incl. external datasets
Limitations	Largely single health system; systemic AUROCs moderate; possible confounding by age
Contribution	Openly released medical foundation model; template for medical MAE

Science reading

- Retina carries systemic information (rungs 1–2)
- Mechanism: hypothesis, not finding

Check before believing

- Does age alone explain the prediction?
- Does it hold across devices and populations?

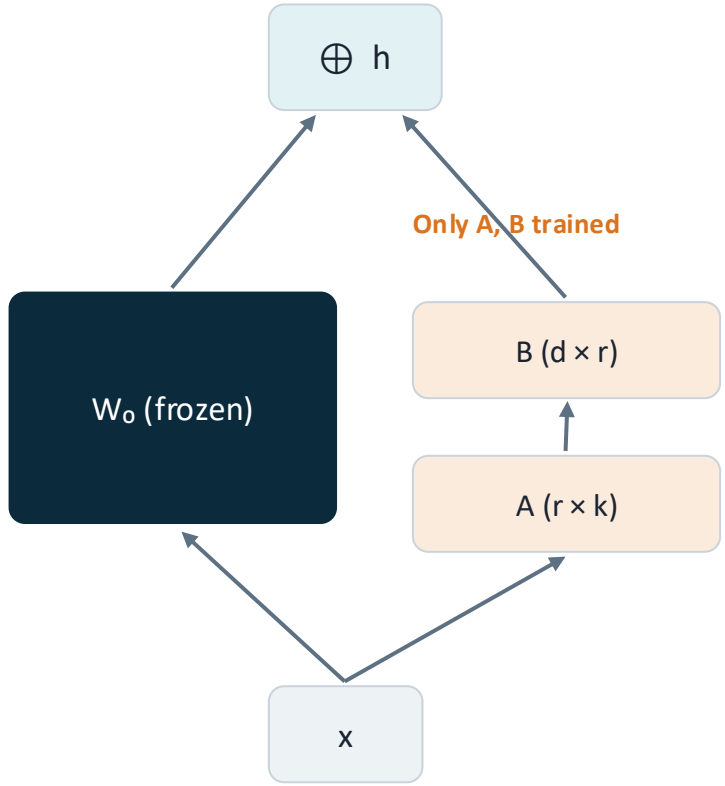
Zhou Y. et al., A foundation model for generalizable disease detection from retinal images, Nature 2023, doi:10.1038/s41586-023-06555-x.

Adapting Foundation Models: Adapters and LoRA

Freeze the backbone; learn a low-rank correction

$$h = W_0 x + (\alpha/r) \cdot B\,A\,x, \quad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times k},\ r \ll \min(d, k)$$

W_0	frozen pretrained weight (d × k)
A, B	trainable low-rank factors; B initialised to 0
r	rank (e.g. 4–16)
α	scaling hyperparameter



Parameter count example

Full update of a 4096 × 4096 projection: 16,777,216 weights
LoRA, r = 8: 2 × 4096 × 8 = 65,536 weights (0.39%)
Merged at inference: $W = W_0 + (\alpha/r)BA \rightarrow$ no extra latency

Hu et al., LoRA: Low-rank adaptation of large language models, ICLR 2022; Houlsby et al., Parameter-efficient transfer learning for NLP, ICML 2019.

Pathology Foundation Models (2024)

Segment · Diagnose · Discover

Gigapixel slides, self-supervision at scale — and the first image-to-genome predictions

Model	Venue	Pretraining scale (approx., as reported)	Distinctive capability	Evidence
UNI	Nat. Med. 2024	>100M tiles, ~100k slides	General-purpose tile encoder, many tasks	Retrospective benchmarks
Virchow	Nat. Med. 2024	~1.5M slides, one cancer centre	Pan-cancer incl. rare cancers	Retrospective incl. external
Prov-GigaPath	Nature 2024	~1.3B tiles, ~171k slides	Slide-level long-sequence encoder; mutations	Retrospective benchmarks
CHIEF	Nature 2024	Tens of thousands of slides, many sites	Detection, origin, genomics, survival	Many independent cohorts

Image → genome

- Morphology predicts molecular alterations
- Hypothesis-generating, not mechanism

Shortcut risks

- Stain, scanner, and site signatures
- Stratify by site; test across scanners

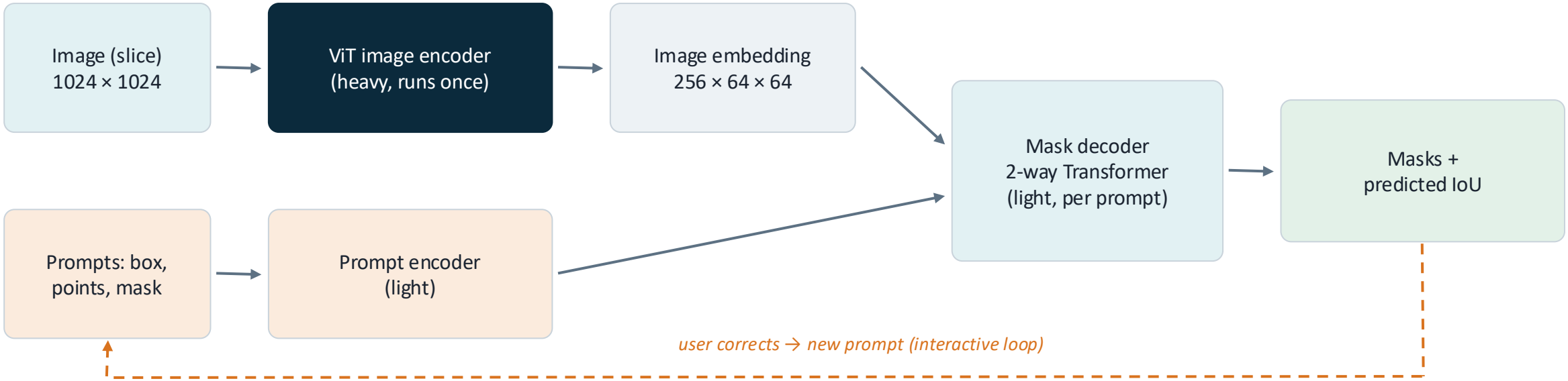
Same ideas as radiology

- Self-supervision, MIL aggregation
- Long-context attention over tiles

Wang X. et al., CHIEF, Nature 2024; Xu H. et al., Prov-GigaPath, Nature 2024; Chen R.J. et al., UNI, Nat. Med. 2024; Vorontsov E. et al., Virchow, Nat. Med. 2024. Scales are approximate as reported.

SAM and MedSAM: Promptable Segmentation

A heavy image encoder runs once; a light prompt encoder and mask decoder run per interaction



MedSAM adaptation

- Fine-tuned on >1.5M medical image–mask pairs, 10 modalities (per paper)
- Bounding-box prompts: less ambiguous than points
- Loss: Dice + binary cross-entropy

Pre-processing

- CT windowing; MRI percentile scaling
- Grey-scale → 3-channel, resize to 1024^2
- 3D handled slice by slice

Interaction cost

- Encoder once per slice
- Each new prompt: decoder only
- Enables real-time correction

Kirillov et al., Segment Anything, ICCV 2023; Ma et al., Segment anything in medical images, Nat. Commun. 2024, doi:10.1038/s41467-024-44824-z.

Paper Analysis: MedSAM (Nat. Commun. 2024)

Natural-image pretraining is a starting point, not a medical segmenter

Element	MedSAM
Scientific problem	Task-specific segmenters do not generalise across modalities and targets
Innovation	Large-scale medical fine-tuning of a promptable SAM-style model
Architecture	ViT image encoder, prompt encoder, mask decoder; box prompts
Training data	>1.5M image–mask pairs, 10 modalities, >30 cancer types (as reported)
Objective	Dice + binary cross-entropy on predicted masks
Experimental design	Internal and external validation tasks; comparisons with SAM and specialist U-Nets
Evidence type	Retrospective benchmarking incl. external datasets — not prospective
Limitations	2D slice-wise; needs human box; small / low-contrast targets; prompt sensitivity

Why SAM alone fails

- Weak, low-contrast boundaries
- Clinically (not visually) defined objects
- Small targets in large fields of view
- Grey-scale, quantitative, 3D data

Contribution

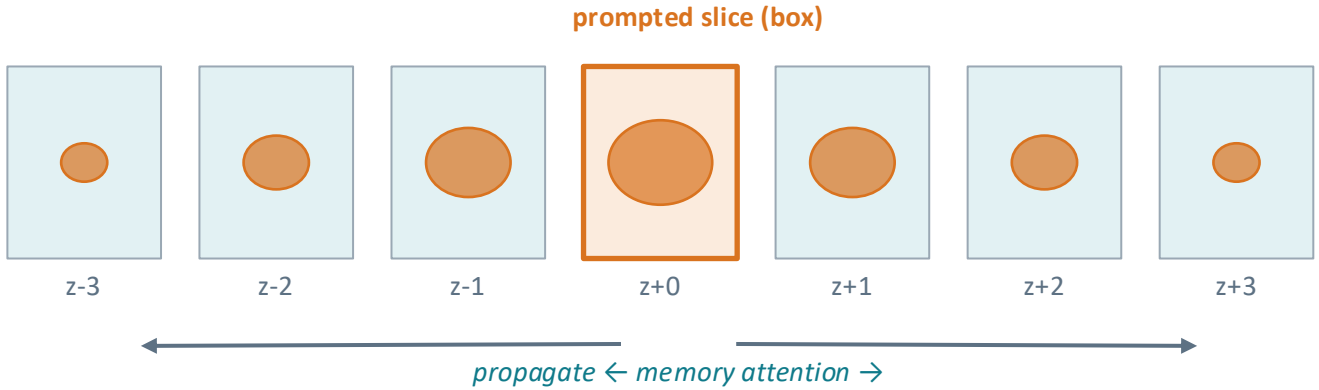
- Promptable segmentation as a reusable medical capability
- Human-in-the-loop by design

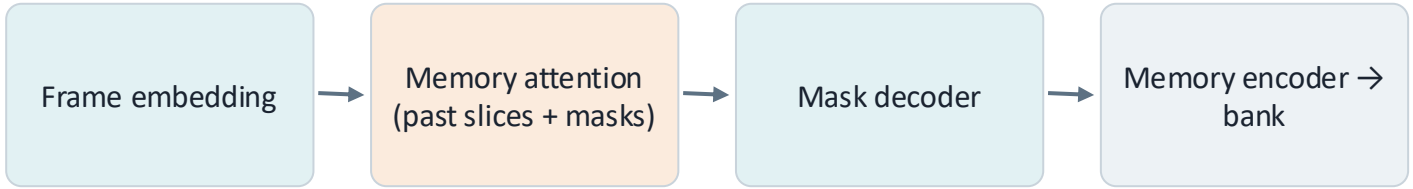
Ma et al., Nat. Commun. 2024, doi:10.1038/s41467-024-44824-z. Details summarised from the paper; verify specifics before teaching.

MedSAM2: Volumes as Ordered Slice Sequences



Treat a 3D volume like a video: propagate masks through a memory of previous slices





Correct a drifting slice → add it to memory → re-propagate (human-in-the-loop).

Gains

- 3D consistency without 3D convolutions
- One prompt → whole volume
- Same model for video (echo, endoscopy)

Failure modes

- Lesions that appear/split between slices
- Small lesions; low-contrast edges
- Thick slices (e.g. 5 mm) break continuity
- Drift far from the prompted slice

Ma et al., MedSAM2: Segment anything in 3D medical images and videos, arXiv:2504.03600 (2025) — preprint; check for a peer-reviewed version. Ravi et al., SAM 2, arXiv:2408.00714 (2024).

Paper Analysis: BiomedParse (Nat. Methods 2025)



From box prompts to text prompts: segment, detect, and recognise objects by name

	MedSAM (2024)	BiomedParse (2025)
Prompt	Bounding box (human)	Text naming the object
Tasks	Segmentation	Segmentation + detection + recognition
Data	>1.5M image–mask pairs	Millions of image–mask–text triples (LLM-harmonised labels)
Strength	Precise when box is good	No box; irregular shapes; whole-image parsing
Failure to test	Low-contrast boundaries	Prompts for absent objects → spurious masks
Evidence	Retrospective incl. external	Retrospective benchmarks

Why it matters for agents

- Language interface = callable tool
- Orchestrator can request ‘segment X’

Test with negatives

- Prompt for objects that are absent
- Measure false-positive mask rate

Zhao T. et al., A foundation model for joint segmentation, detection and recognition of biomedical objects across nine modalities, Nat. Methods 2025. Scale as reported; verify.

Case Study 3: Interactive Liver Lesion Segmentation



Prompt, propagate, correct, measure — and validate against a supervised baseline

1 Box prompt on max-extent slice → 2 Prompt encoding (2 corner tokens) → 3 Mask inference → 4 Propagate ± slices via memory → 5 Manual correction → re-propagate → 6 Volume + QA

Volume (illustrative)

Spacing 0.8 × 0.8 × 3.0 mm
⇒ 1.92 mm^3 per voxel
3,200 lesion voxels
⇒ 6,144 mm^3 ≈ 6.1 cm^3

Report with inter-operator range, not a single number.

	Interactive (MedSAM2-style)	Supervised (nnU-Net)
Labelled training data	None for this task	Required per task
Human effort per case	Prompt + corrections	Review only
Transfer to new lesion type	Immediate	Retrain
Consistency	Operator-dependent	Deterministic
Evaluation	Dice, HD95, #clicks, time, inter-operator spread	Dice, HD95, lesion-wise detection

Failure cases: leakage into vessels; adjacent lesions merged; tiny satellite lesions missed.

Ma et al., Nat. Commun. 2024; Ma et al., arXiv:2504.03600 (MedSAM2, preprint); Isensee et al., Nat. Methods 2021 (nnU-Net baseline).

Generative Models for Medical Imaging



Model p(x) or p(x | c) — then sample, translate, restore, or augment

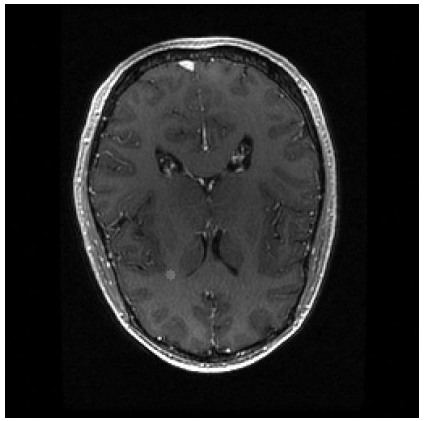
Model	Objective	Strength	Weakness	Medical use
Autoencoder	$\|x - D(E(x))\|^2$	Compression, anomaly scores	No principled sampling	Latent spaces, anomaly detection
VAE	ELBO = $E[\log p(x\|z)] - KL(q\|\|p)$	Probabilistic latent	Blurry samples	Representation, latent priors
GAN	$\min_G \max_D$ adversarial game	Sharp images, fast sampling	Mode collapse, instability	Image translation (pix2pix-type)
Diffusion	Denoising score matching	Fidelity, coverage, stable	Slow sampling	Synthesis, inverse problems
Latent diffusion	Diffusion in autoencoder latent	Scales to 3D volumes	Autoencoder bottleneck errors	3D CT/MRI synthesis (MAISI)

Conditioning c ∈ {another sequence, a segmentation mask, text, raw measurements y} turns a generator into a translation, controllable-synthesis, or reconstruction model.

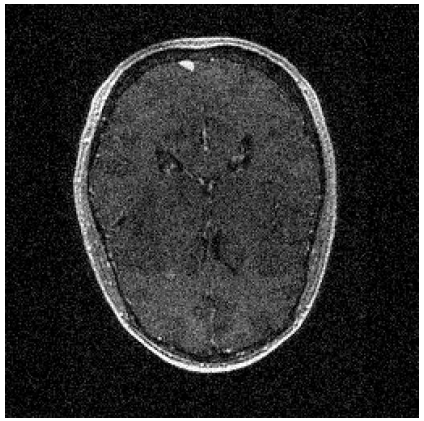
Kingma & Welling, VAE, ICLR 2014; Goodfellow et al., GAN, NeurIPS 2014; Ho et al., DDPM, NeurIPS 2020; Rombach et al., Latent diffusion, CVPR 2022.

Diffusion: The Forward Process

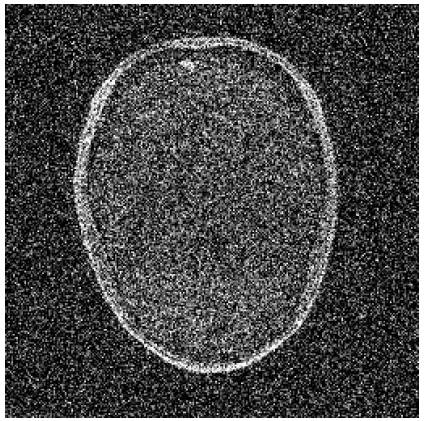
Gradually destroy structure with Gaussian noise; learn to undo it



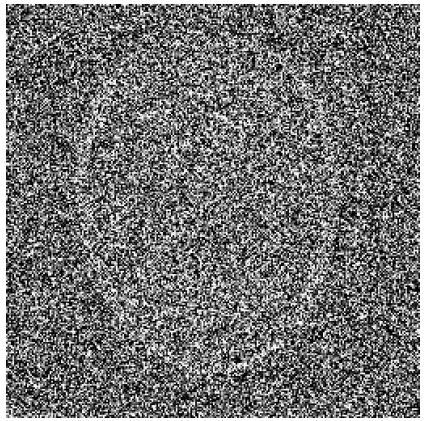
t = 0, $\bar{\alpha}_t = 0.9999$



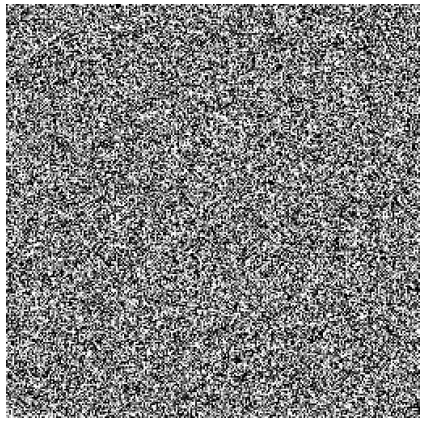
t = 50, $\bar{\alpha}_t = 0.970$



t = 200, $\bar{\alpha}_t = 0.656$



t = 500, $\bar{\alpha}_t = 0.078$



t = 999, $\bar{\alpha}_t = 4\times10^{-5}$

$$q(x_t \mid x_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t)\, I\right), \quad \bar{\alpha}_t = \Pi_{s \le t}\,(1-\beta_s)$$

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{(1-\bar{\alpha}_t)}\, \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

Observation

- Cortex and vessels lost by t ≈ 200; head outline survives to t ≈ 500
- High frequencies destroyed first
- Reverse process: coarse anatomy first, detail last

Ho, Jain & Abbeel, Denoising diffusion probabilistic models, NeurIPS 2020. Panels computed here: linear β schedule $10^{-4} \rightarrow 0.02$, T = 1000, real T1 brain slice (3D Slicer sample data).

Training and Conditioning Diffusion Models



Predict the noise; condition through concatenation, cross-attention, or guidance

$$L_{diffusion} = E_{x_0,\, t,\, \varepsilon}\left[\| \varepsilon - \varepsilon_\theta(x_t, t, c) \|^2 \right]$$

ε	true Gaussian noise added at step t
ε_θ	denoising network (usually a U-Net)
t	timestep, sampled uniformly from 1…T
c	condition: sequence, mask, text, metadata

classifier-free guidance, weight w

$$\tilde{\varepsilon} = (1 + w)\, \varepsilon_\theta(x_t, c) - w\, \varepsilon_\theta(x_t, \varnothing)$$

Reverse step: $x_{t-1} = (1/\sqrt{\alpha_t})(x_t - \beta_t/\sqrt{(1-\bar{\alpha}_t)} \cdot \varepsilon_\theta) + \sigma_t\, z$. Inverse problems: interleave a data-consistency step with y.

Spatial condition

- T1 image, segmentation mask
- Concatenate as input channels
- Or ControlNet-style branch

Non-spatial condition

- Text, scanner, sequence metadata
- Cross-attention: Q from U-Net, K/V from tokens

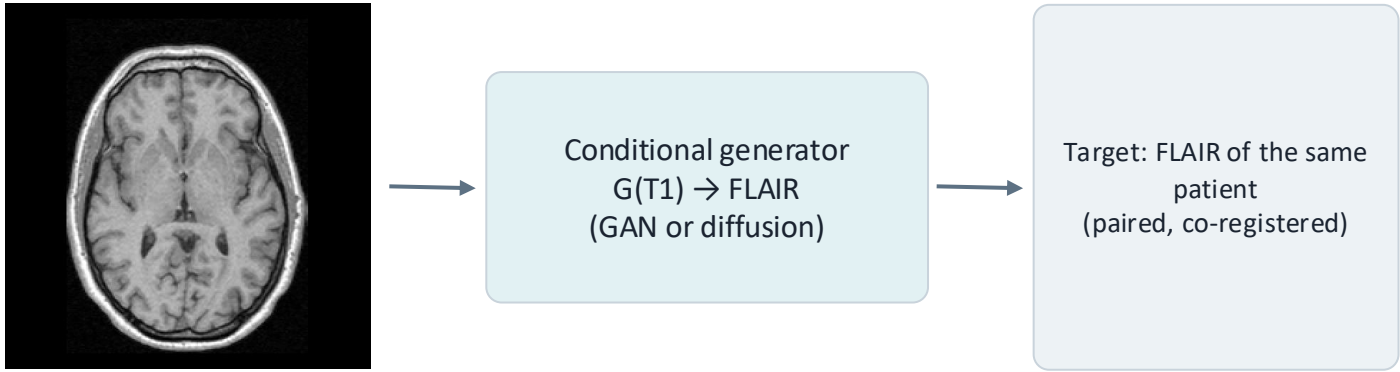
Anatomy preservation

- Guidance ↑ adherence, ↓ diversity
- Check structure with segmentation consistency

Ho et al., NeurIPS 2020; Ho & Salimans, Classifier-free diffusion guidance, NeurIPS Workshop 2021; Rombach et al., CVPR 2022.

Case Study 4: Synthesising FLAIR from T1

Plausible is not correct: synthesis can invent or erase pathology



Failure: plausible but wrong

- Few tumours in training ⇒ lesion erased to look 'normal'
- Tumour-heavy training ⇒ lesion-like signal invented
- Fluent texture hides missing information

$$L = \lambda_1 \|G(T1) - FLAIR\|_1 + \lambda_{adv} L_{adv} \quad (\text{or } L_{diffusion} \text{ with } c = T1)$$

- **Data:** paired, co-registered T1/FLAIR; patient-level split
- **Anatomical consistency:** segment real vs synthetic FLAIR; compare lesion volume
- **Information limit:** oedema and infiltration are weakly visible on T1

Evaluation that matters

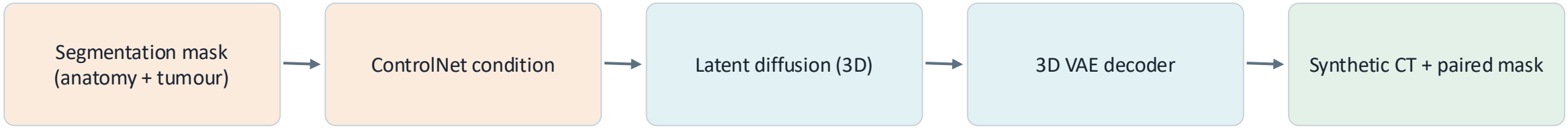
- Lesion detection and volume on synthetic vs real
- Downstream segmentation Dice with synthetic input
- Blinded reader discrimination test
- Never as sole basis for diagnosis

Dar et al., Image synthesis in multi-contrast MRI with conditional GANs, IEEE TMI 2019; Cohen, Luck & Honari, Distribution matching losses can hallucinate features in medical image translation, MICCAI 2018.

3D CT Synthesis and Generative FMs

Enhance

Controllable synthetic volumes for augmentation — validated by downstream utility, not looks



Element	MAISI (WACV 2025)	Generative FM (Nat. Med. 2025)
Problem	Scarce, costly 3D annotated CT	Scarce data across modalities
Approach	3D VAE + latent diffusion + ControlNet on masks	Text-conditioned generative foundation model
Evidence type	Technical feasibility + downstream augmentation benchmarks	Retrospective; check reader / downstream studies in paper
Key limitation	Synthetic texture/noise ≠ scanner physics	Summarise only after reading the full paper

Risks

- Memorisation → privacy leakage
- Distribution mismatch (HU, noise)
- Synthetic pathology may be unrealistic

Evaluate by utility

- Real external test set only
- Δ Dice / detection with vs without synthetic data
- Nearest-neighbour memorisation audit

Guo et al., MAISI: Medical AI for Synthetic Imaging, WACV 2025; Self-improving generative foundation model for synthetic medical image generation and clinical applications, Nat. Med. 2025, doi:10.1038/s41591-024-03359-y.

From Image-Only AI to Vision–Language Models



Five ways to connect images and language — each with different grounding properties

1 Dual encoder

- Image enc. ‖ text enc.
- Contrastive alignment
- Retrieval, zero-shot labels
- Cannot generate text

2 Cross-attention fusion

- Text decoder attends to image features
- Gated cross-attn layers
- Few-shot interleaved input

3 Encoder + projector + LLM

- Visual tokens in LLM space
- Linear / MLP projector
- Instruction tuning
- Most common today

4 Native multimodal

- Joint tokenisation from pretraining
- Single Transformer
- Highest compute

5 3D medical VLM

- Volumetric encoder
- Token aggregation (perceiver, pooling)
- Multi-sequence inputs

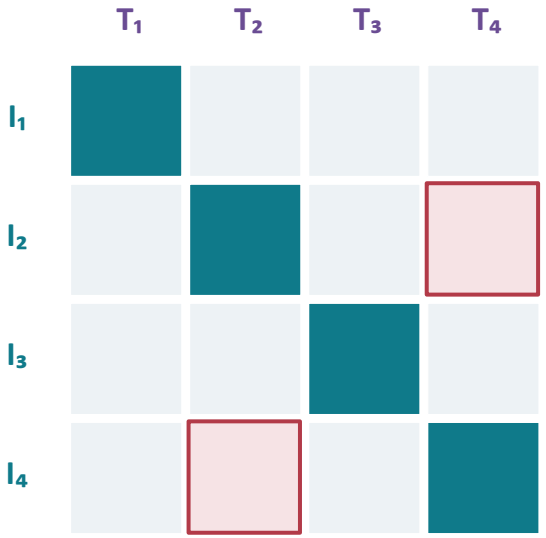
A model that can describe can also hallucinate. Generation capability must come with grounding and verification.

Radford et al., CLIP, ICML 2021; Alayrac et al., Flamingo, NeurIPS 2022; Liu et al., LLaVA, NeurIPS 2023; Li et al., LLaVA-Med, NeurIPS 2023 (Datasets & Benchmarks).

CLIP-Style Image–Text Alignment

Make matched image–report pairs the most similar entries in each row and column

$$L = \tfrac{1}{2}\left[-\frac{1}{N}\sum_i \log\left(e^{S_{ii}} / \sum_j e^{S_{ij}} \right) - \frac{1}{N}\sum_j \log\left(e^{S_{jj}} / \sum_i e^{S_{ij}} \right) \right], \quad S_{ij} = u_i^{T} v_j / \tau$$



Teal: matched pairs (targets). Red border: false negative — I_2 and I_4 both have 'No acute abnormality' reports.

Zero-shot: score prompts 'FLAIR hyperintense lesion' vs 'no abnormality' by similarity.

Image encoder f(·)
Text encoder g(·)
→ shared space

Medical complications

- Many identical normal reports ⇒ false negatives
- Negation ('no evidence of …') is easy to miss
- Report describes study, not slice
- Long text, abbreviations, templates

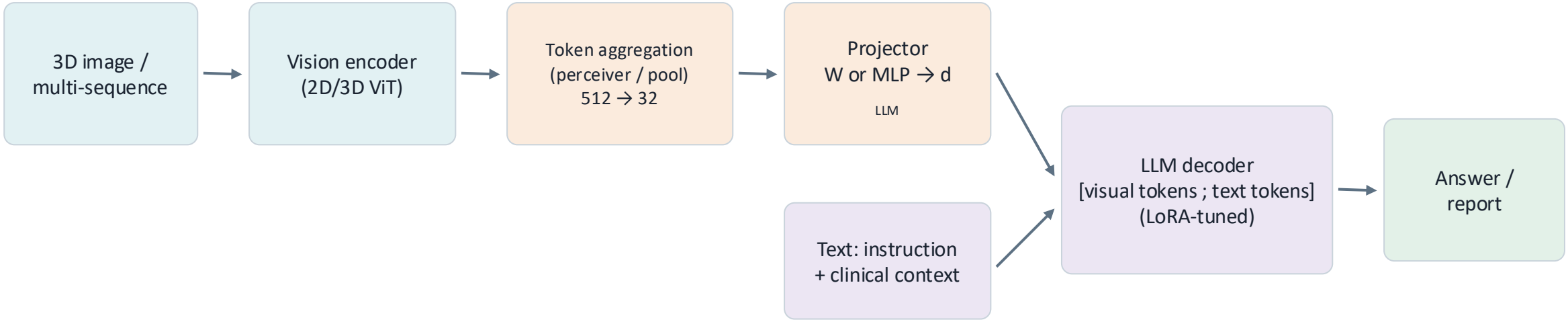
Remedies

- Soft semantic targets (MedCLIP)
- Section-level or sentence-level alignment
- Negation-aware text encoders

Radford et al., ICML 2021; Wang et al., MedCLIP, EMNLP 2022; Tiu et al., Nat. Biomed. Eng. 2022.

Medical VLM Architecture: Encoder, Projector, LLM

Visual features become tokens the language model can attend to



Training stage	Trainable	Data
1 Feature alignment	Projector only	Image–caption / image–report pairs
2 Instruction tuning	Projector + LoRA on LLM (± encoder)	(image, instruction, answer) triples
3 Optional preference tuning	LLM adapters	Clinician-ranked outputs

3D example: 128×128×64 CT, 16×16×8 patches ⇒ 512 tokens; 32 learned queries ⇒ 32 visual tokens. What small findings did the compression discard?

Liu et al., LLaVA, NeurIPS 2023; Alayrac et al., Flamingo, NeurIPS 2022; Hu et al., LoRA, ICLR 2022.

Paper Analyses: RadFM and CT-CLIP / CT-CHAT

Compare designs and evidence, never leaderboard numbers across different datasets

	RadFM (Wu et al., 2025)	CT-CLIP / CT-CHAT (Hamamci et al., 2026)
Scientific problem	Narrow, mostly 2D radiology models	Lack of large open 3D CT–report data
Innovation	Generalist 2D + 3D model with interleaved image–text	Public 3D chest CT–report dataset; 3D CLIP; chat model
Architecture	Visual encoder (2D/3D) + perceiver + LLM	3D ViT encoder + text encoder; encoder + LLM for chat
Training data	Large web-scale 2D/3D medical corpus + radiology subset	Chest CT volumes with reports (tens of thousands)
Objective	Autoregressive LM conditioned on images	Contrastive (CT-CLIP); instruction tuning (CT-CHAT)
Evidence type	Retrospective benchmarks; expert rating	Retrospective; external validation for detection
Limitations	Web data noise; token compression; hallucination	Chest only; non-contrast; report-derived labels
Contribution	Generalist radiology FM direction	Open 3D resource; zero-shot 3D detection

Wu et al., Nat. Commun. 2025, doi:10.1038/s41467-025-62385-7; Hamamci et al., Nat. Biomed. Eng. 2026, doi:10.1038/s41551-025-01599-y. Summaries are high-level; confirm dataset sizes in the papers.

Vision–Language Beyond Radiology



Pathology and echocardiography show the same recipe — and the same evaluation gaps

Model	Venue	Modality	Recipe	Notable result	Caution
CONCH	Nat. Med. 2024	Pathology	Contrastive image–caption	Zero-shot classification, retrieval	Caption quality and coverage
PathChat	Nature 2024	Pathology	Vision encoder + LLM, instruction-tuned	Expert-rated assistant answers	Fluent errors; expert time
EchoCLIP	Nat. Med. 2024	Echo video	CLIP on videos + reports	Zero-shot function, devices	Re-identifies patients: privacy

Same recipe as the radiology VLMs above. Same gaps: benchmarks and expert ratings, single-institution data, and new privacy risks.

Discussion: if a model can recognise the same patient across echo studies, what else could it infer that the patient did not consent to share?

Lu M.Y. et al., CONCH, Nat. Med. 2024; Lu M.Y. et al., PathChat, Nature 2024; Christensen M. et al., EchoCLIP, Nat. Med. 2024. Scales approximate as reported.

Captioning Is Not Grounded Diagnosis



A diagnosis requires localisation, measurement, context, and verifiable evidence

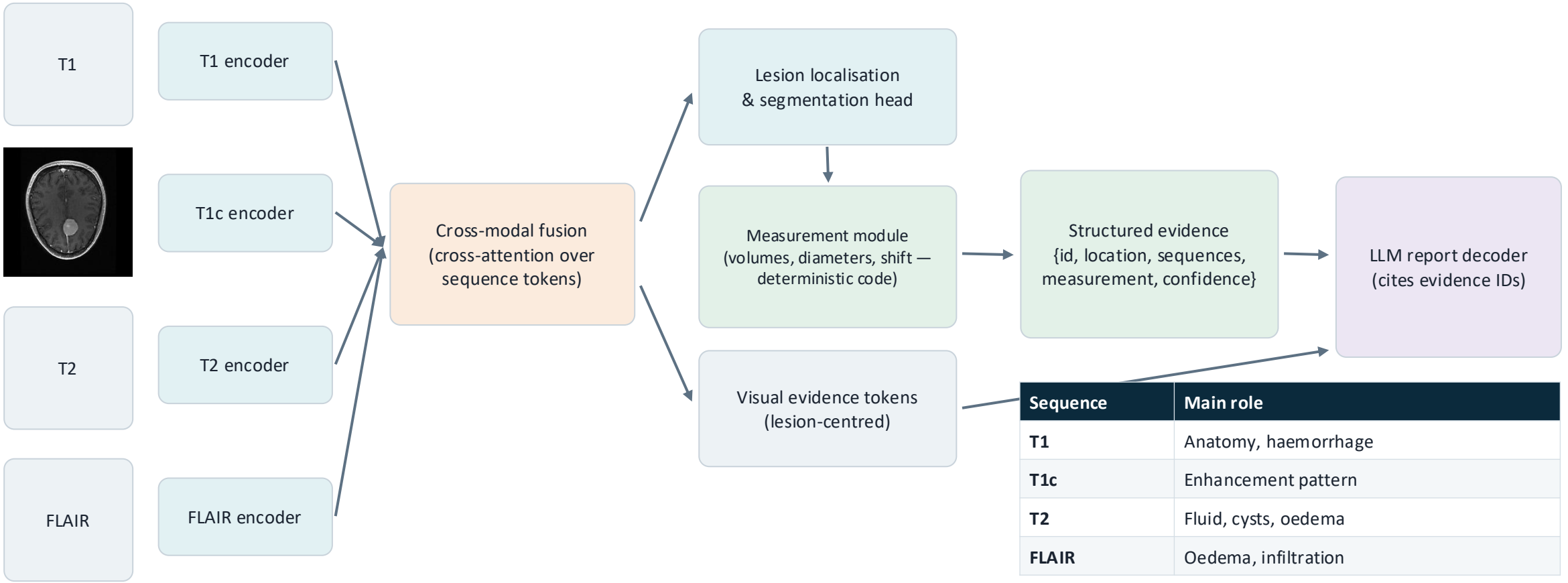
Requirement	Image captioning	Clinically grounded diagnosis
Objective	Likely text for typical images	Correct findings for this patient
Localisation	Implicit or absent	Anatomical location, laterality, slice
Quantification	Rare	Size, volume, change vs prior with units
Context	Image only	History, priors, labs, indication
Uncertainty	Fluent regardless	Hedged, differential, abstain when needed
Verifiability	No link to evidence	Each claim traceable to image evidence

Test for every generated sentence: can you point to the voxels and the measurement that support it? If not, it is a caption, not a finding.

Case Study 5: Brain MRI to Structured Report

Segment · Diagnose · Report

Separate perception, measurement, and language — and keep the evidence between them explicit



Sequence	Main role
T1	Anatomy, haemorrhage
T1c	Enhancement pattern
T2	Fluid, cysts, oedema
FLAIR	Oedema, infiltration

Design synthesises ideas from RadFM, BrainGPT (brain CT), and segmentation literature; this is a teaching architecture, not a published system.

Case Study 5: Report and Evidence Table

Every statement links to sequences, locations, and measurements

ILLUSTRATIVE VALUES — not experimental results

EXAMINATION MRI brain without and with gadolinium (fictional case).

TECHNIQUE Axial T1, T1 post-contrast, T2, FLAIR; resampled to 1 mm isotropic.

FINDINGS [E1] Rim-enhancing left frontal lesion, 3.1 cm maximal diameter, [E2] with central non-enhancing component. [E3] Surrounding T2/FLAIR hyperintensity. [E4] Mild local mass effect; midline shift 1.4 mm. [E5] No additional enhancing lesion identified.

IMPRESSION Solitary rim-enhancing left frontal mass with surrounding signal abnormality; differential includes high-grade glial neoplasm and metastasis. PROVISIONAL — requires radiologist review.

ID	Claim	Sequences	Location	Model observation	Conf.
E1	Rim enhancement	T1c vs T1	L frontal, z 88–118	ET 4.2 cm^3; ring pattern	0.91
E2	Central non-enhancing	T1c, T2	Lesion core	NCR 2.1 cm^3; T2-bright	0.84
E3	Surrounding hyperintensity	FLAIR, T2	Peri-lesional	ED 18.7 cm^3	0.88
E4	Midline shift 1.4 mm	T1	Septum pellucidum	Shift 1.4 mm (threshold-near)	0.76
E5	No other enhancing lesion	T1c	Whole brain	No ET component > 0.1 cm^3	0.69

Whole tumour = 4.2 + 2.1 + 18.7 = 25.0 cm^3 (consistent with Case Study 2). Absence claims (E5) carry lower confidence.

Fictional case and values; not patient data.

Autoregressive Report Generation



A report is a sequence of tokens, each conditioned on the images, the context, and all previous tokens

$$P(R \mid I, C) = \Pi_{t=1}^{T} P(r_t \mid r_{<t}, I, C)$$

$R = (r_1 \ldots r_T)$	report tokens
I	images (all sequences)
C	clinical context: indication, history, priors
$r_{<t}$	previously generated tokens

teacher forcing

$$L_{CE} = -\Sigma_t \log P(r^*_t \mid r^*_{<t}, I, C)$$

Exposure bias: at inference the model conditions on its own prefix — an early ‘Small …’ instead of ‘No …’ steers every later token.

Instruction tuning

- (images, instruction, report) triples
- Teaches format and section structure

Retrieval augmentation

- Adds similar reports or guidelines to C
- Risk: copies findings from the retrieved case

Structured reporting

- Template fields per region / finding
- Easier to verify and evaluate

Williams & Zipser, Neural Comput. 1989 (teacher forcing); Lewis et al., Retrieval-augmented generation, NeurIPS 2020.

Evaluating Reports: Wording versus Findings



Text-overlap metrics measure wording; a minor linguistic difference can be a major clinical error

Metric	Measures	Blind spot
BLEU / ROUGE-L	n-gram precision / LCS F-measure	Negation, laterality, synonyms
BERTScore	Contextual-embedding similarity	Opposite findings embed closely
Clinical entity F1	Agreement of extracted findings	Labeller errors; fixed label set
RadGraph F1	Entities + relations	Extraction model quality
Clinician review	Errors by severity, acceptability	Cost, reader variability

	Reference	Generated	ROUGE-L F
A	No evidence of intracranial hemorrhage	Small intracranial hemorrhage is present	0.40
B	No acute intracranial hemorrhage	Acute intracranial hemorrhage	0.857

Both pairs: opposite clinical meaning. Pair B scores high. Hand-computed, lower-cased whitespace tokens.

Omission
Present finding not reported

Hallucination
Absent finding asserted

Laterality
Left / right swapped

Contradiction
Findings ≠ impression

Measurement
Wrong size, units, change

Missed urgent
Critical result not flagged

Weight errors by clinical severity, report at least one clinical metric, and review a stratified sample with experts.

Papineni et al., BLEU, ACL 2002; Lin, ROUGE, 2004; Zhang et al., BERTScore, ICLR 2020; Jain et al., RadGraph, NeurIPS D&B 2021; Yu et al., Patterns 2023.

Paper Analyses: Flamingo-CXR and BrainGPT



Expert evaluation and clinical metrics reveal what text overlap cannot

	Flamingo-CXR (Nat. Med. 2025)	BrainGPT (Nat. Commun. 2025)
Problem	Are AI chest X-ray reports clinically acceptable?	3D brain CT reporting; inadequate NLG metrics
Innovation	Expert error-based evaluation + clinician–AI collaboration	Clinical visual instruction tuning; FORTE metric
Architecture	Flamingo-style VLM, fine-tuned	Multimodal LLM with 3D CT input
Data	Chest X-ray report datasets (multiple sites)	Curated 3D brain CT–report dataset
Evaluation	Radiologist preference + error severity, AI and human	FORTE + traditional metrics + physician review
Evidence type	Expert reader evaluation (retrospective)	Retrospective benchmark + physician assessment
Limitations	2D CXR; setting-dependent; not prospective	CT only; single-centre-type data; not prospective
Contribution	Human–AI collaboration as the evaluation unit	Clinically oriented report metrics

Collaboration between clinicians and vision–language models in radiology report generation, Nat. Med. 2025, doi:10.1038/s41591-024-03302-1; Towards a holistic framework for multimodal LLM in 3D brain CT radiology report generation, Nat. Commun. 2025, doi:10.1038/s41467-025-57426-0.

Trustworthy Multimodal Systems



Language fluency does not establish clinical correctness

Predictive uncertainty

Per-finding probabilities, not one global score

Calibration

Probabilities match observed frequencies on external data

Evidence consistency

Each claim supported by an image-derived observation

OOD detection

Flag new scanners, paediatric, post-surgical inputs

Abstention

Decline when uncertainty or OOD score exceeds threshold

Independent verification

Separate checker with different failure modes — argued, not assumed

Fluent text is produced whether or not evidence exists. Readers anchor on fluent, confident language.

Geifman & El-Yaniv, Selective classification for deep neural networks, NeurIPS 2017; Hendrycks & Gimpel, Baseline for detecting misclassified and OOD examples, ICLR 2017.

Module II Synthesis and Transition



From reusable representations to coordinated reasoning

Self-supervision

Learn from unlabelled scans; learned features are candidate phenotypes, not yet measurements

Foundation models

Promptable, adaptable (LoRA); validate per task

Generative models

Powerful priors; plausible ≠ correct

Vision–language

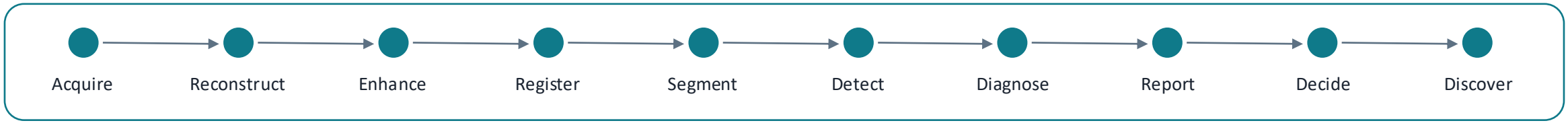
Grounding, structured evidence, clinical metrics

Foundation models can understand and generate multimodal content. Agentic systems extend these capabilities by coordinating specialized tools, evaluating evidence, and managing multistep workflows.

Bridging question: with a segmentation model, a VLM, and a measurement library — who decides what to call, in which order, and what happens when they disagree?

Module III — Agentic AI, Trustworthy Medical Intelligence, and the Future of Clinical Imaging

Tool use · multi-agent orchestration · uncertainty · verification · hybrid models · experimental design · scientific discovery · end-to-end case study



From Foundation Models to Agentic AI

Diagnose · Report · Decide

Each step adds capability — and a new place where errors can enter

Task-specific AI	Foundation models	Multimodal reasoning	Tool-using AI	Multi-agent systems
One model → one output	Pretrain once, adapt many	Images + text → answers	Model calls segmenters, calculators, retrieval	Specialists + orchestrator exchange messages
New risk: Domain shift	New risk: Prompt / adaptation sensitivity	New risk: Ungrounded fluent text	New risk: Wrong tool, wrong arguments	New risk: Correlated errors, propagation

Agentic ≠ more accurate. Agentic = more flexible control flow. Flexibility needs constraints, provenance, and verification.

Yao et al., ReAct, ICLR 2023; Schick et al., Toolformer, NeurIPS 2023; Moor et al., Nature 2023.

Vocabulary of Agentic Systems



Precise terms prevent vague claims

AI agent

Observes state, selects actions by a policy, updates from results, toward a goal

Planning

Decompose goal into ordered sub-tasks; revise as results arrive

Tool invocation

Typed function call; output returned as an observation; tools can be deterministic

Memory

Short-term case state; long-term retrieved guidelines and priors

Orchestration

Controller choosing which agent/tool runs next and when to stop

Feedback

Intermediate results alter subsequent actions

Verification

Check outputs against independent evidence or hard constraints

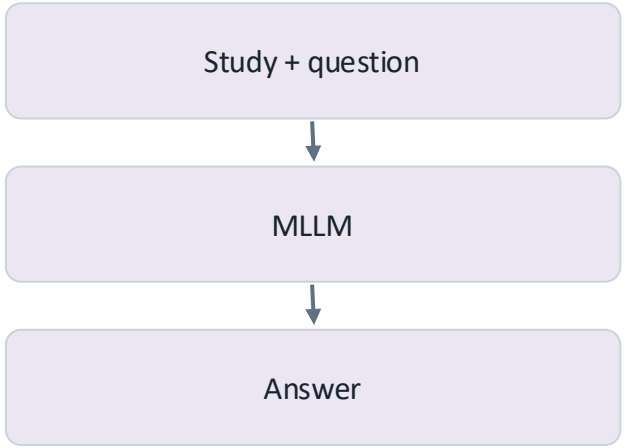
Human-in-the-loop

Defined approval points; diagnosis and accountability remain human

One Answer, a Pipeline, or Orchestration?

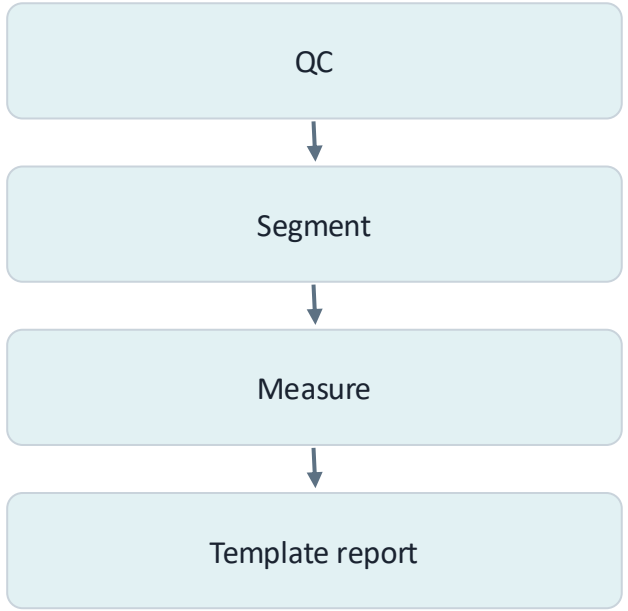
The difference is who decides the control flow — and on what evidence

A Single multimodal model



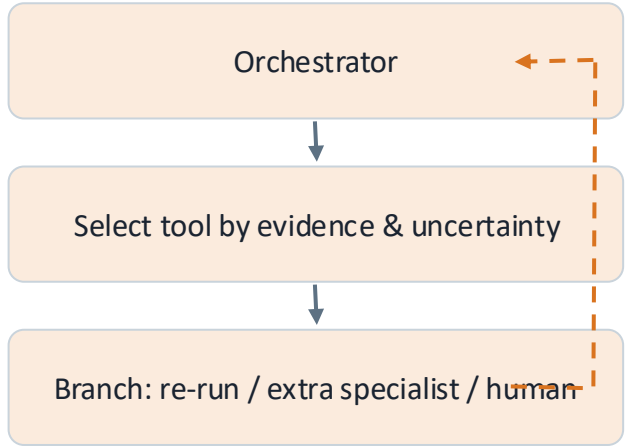
One call; nothing intermediate to audit

B Fixed tool workflow



Predictable, testable; cannot adapt (e.g. missing FLAIR)

C Dynamic orchestration



Adaptive; validation must cover policies, not just models

A Decision-Theoretic Formulation

Choose the action with the highest expected utility — subject to clinical risk constraints

$$a^*_t = \text{argmax}_{a \in \mathcal{A}(s_t)} \; E[\, U(a \mid s_t) \,] \quad \text{s.t.} \quad P(\text{critical miss} \mid \pi) \le \delta, \; \text{hard rules}$$

s_t	state: images, QC flags, findings + uncertainties, context, budget
$\mathcal{A}(s_t)$	allowed actions: tool call, query agent, human review, abstain, finalise
$\pi(a\|s)$	orchestration policy
$U(a\|s)$	utility = information gain − cost − λ·risk
δ, λ	risk bound and risk weight — set clinically, not tuned for accuracy

ILLUSTRATIVE VALUES — not experimental results

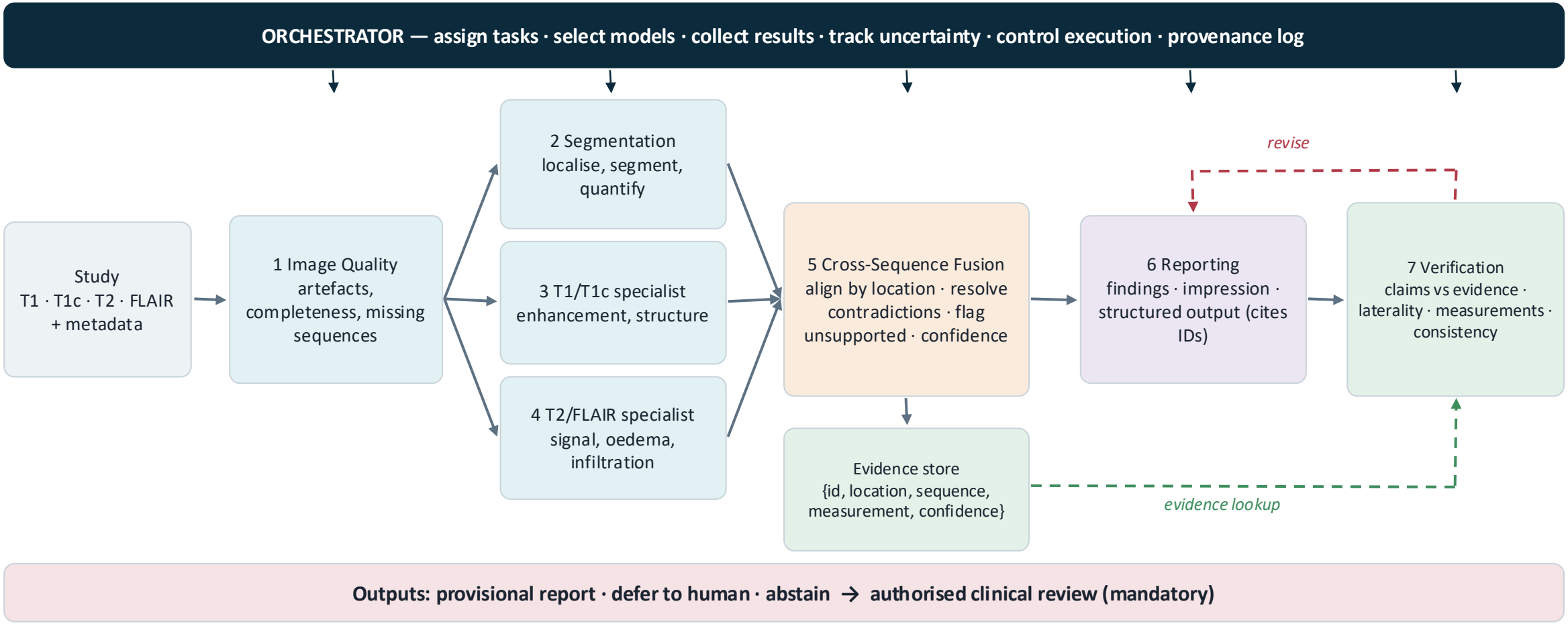
Action (FLAIR degraded)	Gain	Cost	Risk	U (λ=10)	U (λ=1)
a_1 segment on degraded FLAIR	0.30	0.05	0.04	−0.15	0.21
a_2 T2 specialist + flag review	0.25	0.15	0.01	0.00	0.09

λ = 10 selects a_2; λ = 1 selects a_1. The risk weight is a clinical and ethical choice.

Hard rules sit outside the optimisation: never finalise unverified; never auto-clear an out-of-distribution study.

Multi-Agent Brain MRI Architecture

Specialists produce evidence; fusion reconciles it; reporting writes; verification checks — the orchestrator tracks everything



Conceptual architecture for teaching. Not a validated or deployed system.

Agent Roles at a Glance



Typed inputs, auditable tools, characteristic failures — and agents sharing a backbone fail together

Agent	Inputs	Tools → outputs	Characteristic failure
1 Image quality	Headers, all sequences	Header parsing, SNR, artefact classifier → QC scores ± uncertainty, flags	Subtle motion missed; novel protocol mislabelled
2 Segmentation	QC-approved, co-registered sequences	Validated 3D segmenter + deterministic measurement → masks, volumes, uncertainty	Missed small lesions; silent degradation under shift
3 T1/T1c specialist	T1, T1c, masks	Pre/post comparison → enhancement pattern, structure	Intrinsic T1 brightness read as enhancement
4 T2/FLAIR specialist	T2, FLAIR, masks	Extent, oedema vs infiltration, CSF suppression check	Poor CSF suppression mimics lesion
5 Fusion	All observations	Align by overlap/anatomy; resolve or flag contradictions	Correlated errors → confident consensus on a wrong answer
6 Reporting	Evidence store + context	Constrained LLM citing evidence IDs; numbers copied	Fluent unsupported text; findings copied from retrieval
7 Verification	Draft + evidence	Atomic claim checks → approve · revise · defer · abstain	Cannot detect wrong evidence; blind if same model as writer

The Orchestrator and Provenance

If you cannot reconstruct why a sentence was written, you cannot audit it

Responsibilities

- Assign tasks; select validated model versions
- Maintain typed case state
- Propagate uncertainty between stages
- Enforce hard rules, budgets, timeouts
- Append-only provenance log


```
{ "evidence_id": "E1",
  "agent": "T1/T1c specialist",
  "model_version": "t1c-spec v2.3",
  "inputs": ["T1#a91f", "T1c#77c2", "mask#3be0"],
  "observation": "rim enhancement",
  "measurement": {"ET_volume": 4.2, "unit": "cm3"},
  "location": {"region": "left frontal lobe",
     "centroid_mm": [32.0, 41.5, 18.0],
     "slices": [88, 118]},
  "confidence": 0.91, "calibrated": true,
  "verification": "pending",
  "timestamp": "2026-11-08T10:42:13Z" }
```

Orchestration Pseudocode



Twelve steps, three exits: provisional report, deferral, or abstention

```
Input: multisequence MRI study S, context C
 1  q ← QualityAgent.validate(S)
    if q.missing_essential: return DEFER("re-acquire")
 2  obs ← {T1c: SpecA(S), FLAIR: SpecB(S)}
 3  M ← SegAgent.segment(S, q)
 4  m ← measure(M, spacing(S))        # deterministic
 5  E ← collect(obs, M, m) → evidence store
 6  F ← Fusion.check(E)
    if F.unresolved: call extra specialist (≤ k tries)
 7  ctx ← retrieve(C, priors)
 8  D ← Reporter.draft(E, ctx)         # cites IDs
 9  V ← Verifier.check(D, E)           # mandatory
10  flag V.unsupported; revise D (≤ r rounds)
11  if risk(V, F, q) > τ: return ABSTAIN or DEFER
    else: out ← PROVISIONAL(D, E, provenance)
12  require authorised clinical review(out)
```

Exits

- PROVISIONAL report + evidence + provenance
- DEFER to human with reason
- ABSTAIN when risk exceeds τ

Guards

- Bounded retries k, revisions r
- Verification cannot be skipped
- Human review always required

What Exists, What Is Proposed



Distinguish verified implemented systems from conceptual architectures

Component	Evidence status (typical)	Teaching note
Segmentation (e.g. nnU-Net)	Retrospective + some external validation	Mature component
Artefact / quality classifiers	Task-specific retrospective studies	Narrow but useful
Measurement from masks	Deterministic; validated with phantoms	Auditable
VLM report drafting	Benchmarks + expert reader studies	Emerging
Multi-agent orchestration (imaging)	Prototypes, reviews, text benchmarks	Largely conceptual
This lecture's 7-agent system	None — teaching proposal	Conceptual

New risks from agents

- Error propagation across stages
- Correlated errors (shared backbone/data)
- Agreement ≠ correctness
- Prompt injection via retrieved text
- Version and governance complexity

Yi et al., Agentic artificial intelligence in radiology workflow: from image interpretation to report quality control, Front. Med. 2026, doi:10.3389/fmed.2026.1927284 (review — read before citing specifics); Kim et al., NeurIPS 2024; Tang et al., ACL Findings 2024.

Prospective Evidence: AI in Screening

Detect · Decide

MASAI (Lancet Oncology 2023) and PRAIM (Nature Medicine 2025) — what deployment-level evidence looks like

	MASAI (Lancet Oncol. 2023)	PRAIM (Nat. Med. 2025)
Design	Randomised controlled trial, Sweden	Nationwide real-world observational, Germany
Population	>80,000 women (as reported)	>460,000 women (as reported)
Intervention	AI triage to single/double reading + flagged findings	Radiologist-chosen AI-supported reading
Reported result	≈20% more cancers detected; ≈44% less reading workload	Higher detection rate (≈17–18% relative); recall not increased
Key limitation	Interval-cancer endpoint needs follow-up	Self-selection of AI use; observational
Evidence rung	5 — randomised intervention	5 — real-world implementation

What makes it strong

- Narrow task, pre-defined endpoint
- Real workflow, real population

Still open

- Overdiagnosis vs benefit
- Interval cancers, mortality
- Transfer to other programmes

Lång K. et al., MASAI randomised trial, Lancet Oncol. 2023; Eisemann N. et al., nationwide real-world AI implementation in mammography screening (PRAIM), Nat. Med. 2025. Figures as reported; verify before quoting.

Seven Dimensions of Trustworthy AI

Cross-cutting

Complementary, not interchangeable — each can fail while the others look fine



Failure while others look fine	Example
Accurate but miscalibrated	AUROC 0.93, yet 0.9 confidence correct only 70% of the time
Calibrated but unfaithful	Correct probabilities; report text asserts an unmeasured finding
Faithful but wrong	Report matches evidence; evidence came from a misregistered sequence

Gallifant et al., TRIPOD-LLM reporting guideline, Nat. Med. 2025, doi:10.1038/s41591-024-03425-5; CLAIM 2024 Update, Radiol. Artif. Intell., doi:10.1148/ryai.240300.

Aleatoric and Epistemic Uncertainty

Same confidence, different causes — and different actions

$$H(p) = -\Sigma_c\, p_c \log p_c$$

total = aleatoric + epistemic

$$H(\bar{p}) = E_m[\, H(p_m)\,] + I(y; m)$$

Binary, p = 0.62: H = 0.664 nats (0.958 bits); max ln 2 = 0.693

Ensemble members p(lesion)	Mean	Total H	Expected H (aleatoric)	MI (epistemic)
0.90, 0.30, 0.66 (disagree)	0.62	0.664	0.526	0.138
0.62, 0.62, 0.62 (agree)	0.62	0.664	0.664	0.000

Entropies in nats, computed for this slide.

Aleatoric

- Data noise: partial volume, low SNR, ambiguous edges
- Not reduced by more data
- Action: better acquisition or accept ambiguity

Epistemic

- Model ignorance: rare disease, new scanner
- Reduced by relevant data
- Action: defer, OOD check, collect data

When entropy misleads

- Miscalibrated model
- OOD input: confidently wrong (low H)
- Ambiguous label definitions

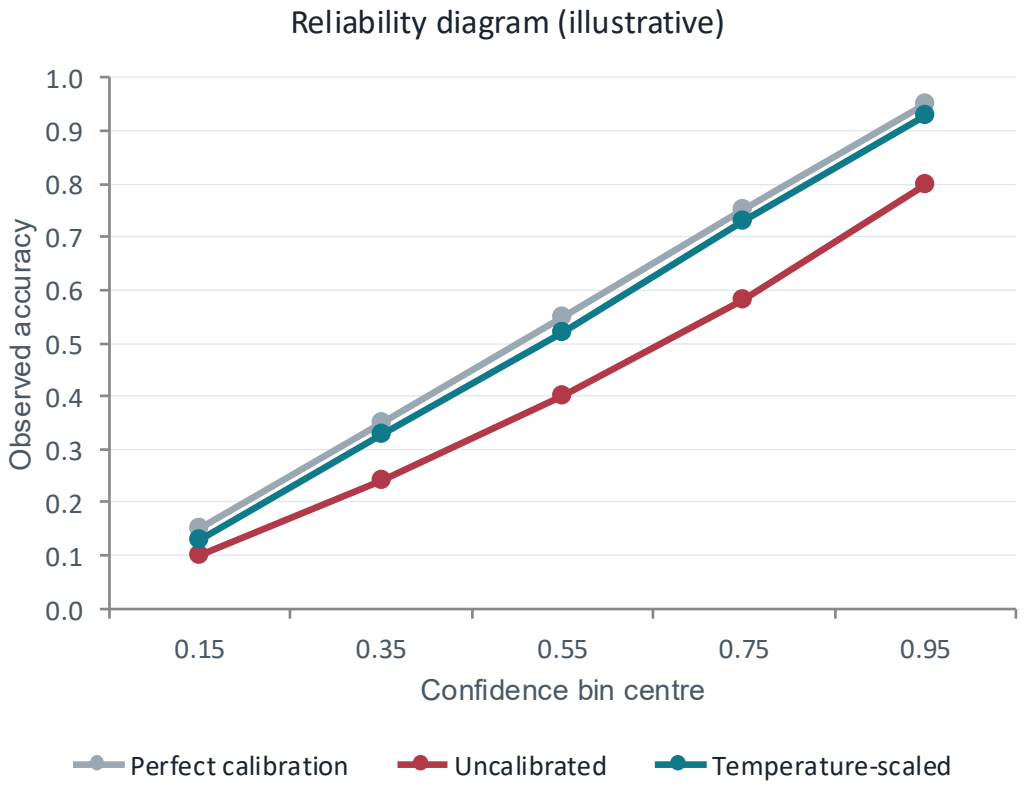
Gal & Ghahramani, MC dropout, ICML 2016; Lakshminarayanan et al., Deep ensembles, NeurIPS 2017; Kendall & Gal, NeurIPS 2017.

Calibration, Conformal Sets, Abstention

Diagnose · Decide

Make scores mean what they say — or return sets and abstentions with guarantees

ILLUSTRATIVE VALUES — not experimental results



$$\mathrm{ECE} = \Sigma_b \, (|B_b| \,/\, n) \cdot |\, \mathrm{acc}(B_b) - \mathrm{conf}(B_b) \,|$$

Illustrative, equal bin weights: ECE 0.126 → 0.022 after temperature scaling.

split conformal

$$C(x) = \{\, y : s(x, y) \le \hat{q} \,\}, \quad P(Y \in C(X)) \ge 1 - \alpha$$

$\hat{q}$ = $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of calibration scores; n = 500, α = 0.1 ⇒ 451st smallest score. Requires exchangeability.

Set {lesion, no lesion}

An honest 'I don't know' → route to a human

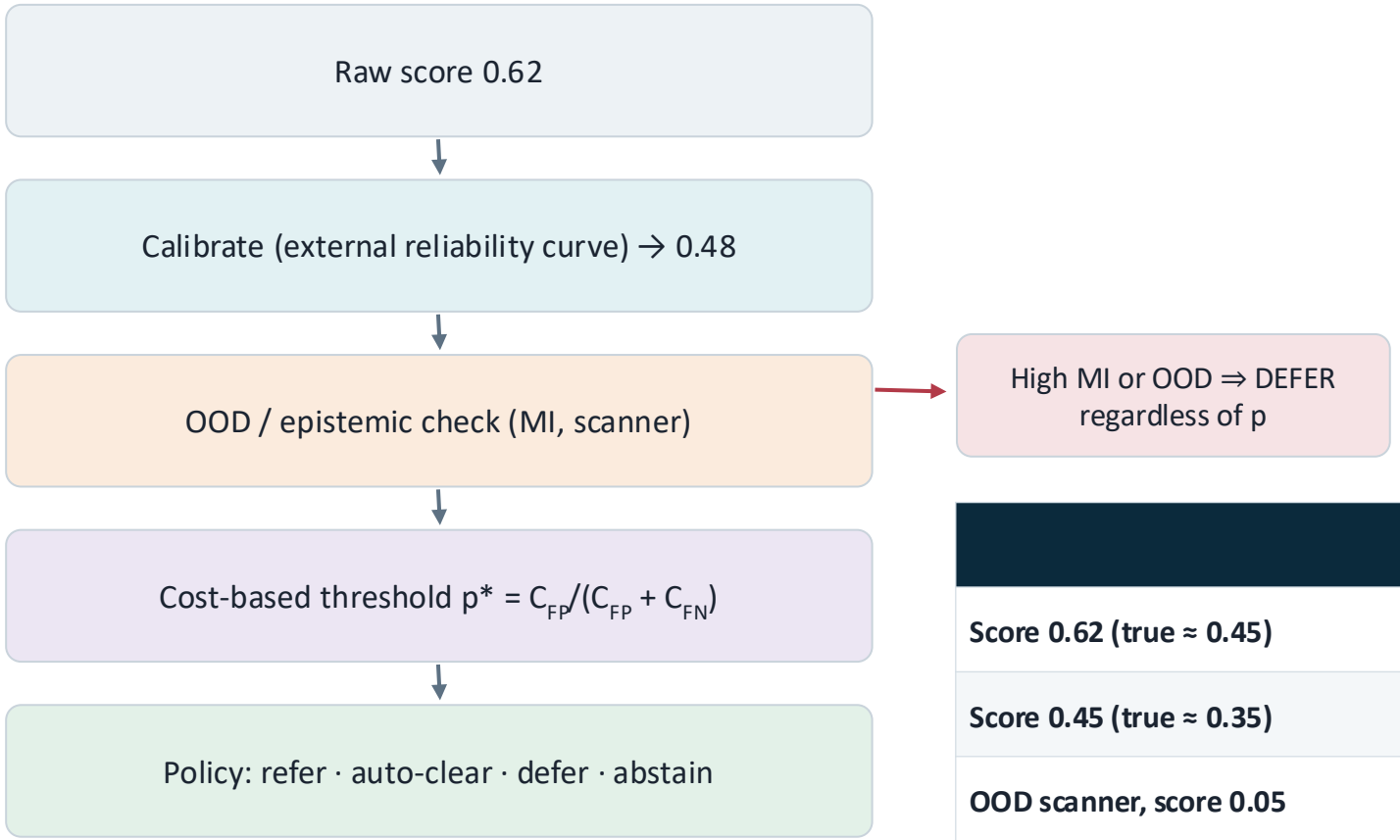
Selective classification

Abstain below threshold; report risk–coverage curves

Guo et al., ICML 2017; Angelopoulos & Bates, A gentle introduction to conformal prediction, arXiv:2107.07511; Geifman & El-Yaniv, NeurIPS 2017.

Worked Example: Referral at Score 0.62

A score is not a decision: calibration, costs, patient risk, and policy turn it into one



ILLUSTRATIVE VALUES — not experimental results

Setting	$C_{FN} : C_{FP}$	p*	Decision at 0.48
Critical finding	20 : 1	0.048	Refer
Screening	2 : 1	0.33	Refer
Auto-clear normals	requires p < 0.02 + low MI	—	Not eligible

	Uncalibrated, t = 0.5	Calibrated + policy
Score 0.62 (true ≈ 0.45)	Refers (by luck)	Refers (by design)
Score 0.45 (true ≈ 0.35)	Does not refer	Refers (p > p*)
OOD scanner, score 0.05	Auto-clears	Defers

Evidence-Faithful Reporting in Practice

Represent the report as claims; check each claim against image-derived evidence

Draft report → Claim extraction (atomic claims) → Match to evidence store → Check: existence, location, laterality, value, negation → Verdict per claim

Fictional case.

Claim	Sequence	Location (evidence)	Model observation	Conf.	Verification
Rim-enhancing lesion, right frontal	T1c vs T1	Centroid +32 mm (patient LEFT); atlas: left frontal	E1: rim enhancement	0.91	**CONTRADICTED — laterality → revise**
Enhancing volume 4.2 cm³	T1c	Left frontal	E1: ET 4.2 cm³	0.91	**SUPPORTED**
Mild hydrocephalus	—	—	No evidence record	—	**UNSUPPORTED → remove**
Midline shift 1.4 mm	T1	Septum pellucidum	E4: 1.4 mm (near 2 mm threshold)	0.76	**SUPPORTED, low conf. → flag**

Limit: the verifier only detects contradictions with the evidence. If evidence is wrong (e.g. flipped orientation in the header), the check passes — design independent checks.

Hybrid Mechanistic–Learned Models

Physics where we trust it, learning where we do not — and a 'digital twin' only if its predictions are validated

physics-informed loss: data + PDE residual

$$L(\theta) = \Sigma_i \, \|u_\theta(x_i, t_i) - d_i\|^2 \; + \; \lambda \, \Sigma_j \, \|\mathcal{N}[u_\theta](x_j, t_j)\|^2$$

Physics-informed networks

- PDE residual as a loss term
- Sparse data, known equations

Neural operators

- Learn the solution operator
- Fast surrogates (weather, fluids)

Differentiable simulators

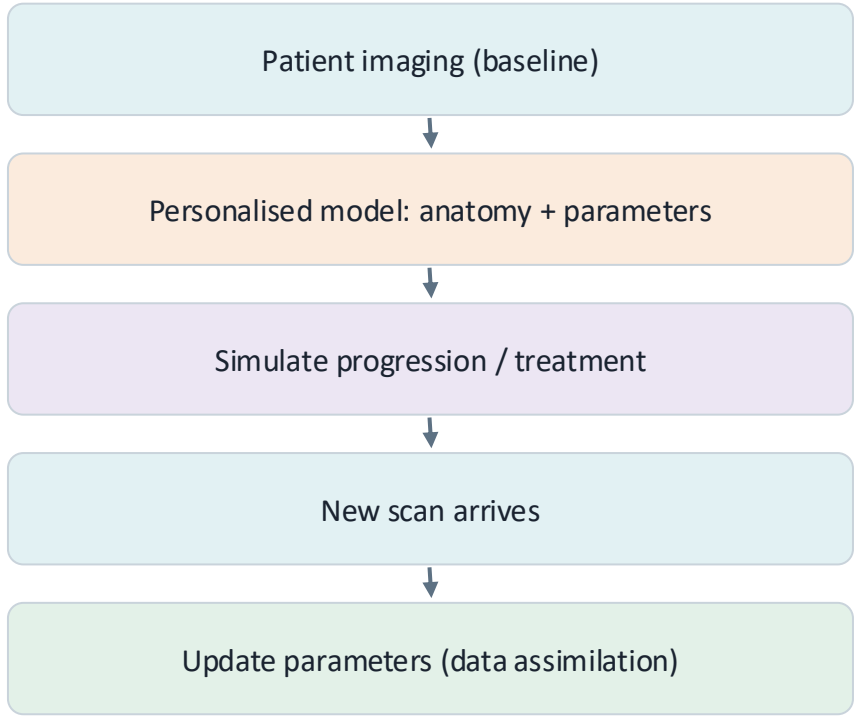
- Bloch simulation in the graph
- Gradients to tissue/sequence params

Simulation-based inference

- Network trained on simulations
- Returns a parameter posterior

Right mechanism → less data, better extrapolation. Wrong or incomplete mechanism → confidently wrong predictions.

Digital twin loop



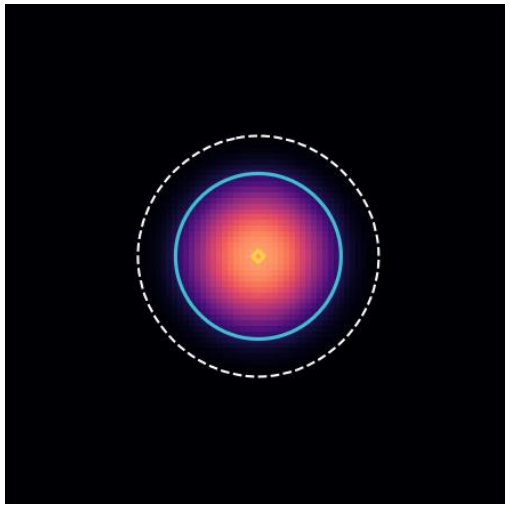
Clinically validated twins are rare: predictions themselves need prospective validation.

Raissi, Perdikaris & Karniadakis, Physics-informed neural networks, J. Comput. Phys. 2019; Li et al., Fourier neural operator, ICLR 2021; Cranmer, Brehmer & Louppe, The frontier of simulation-based inference, PNAS 2020; Corral-Acero et al., Eur. Heart J. 2020.

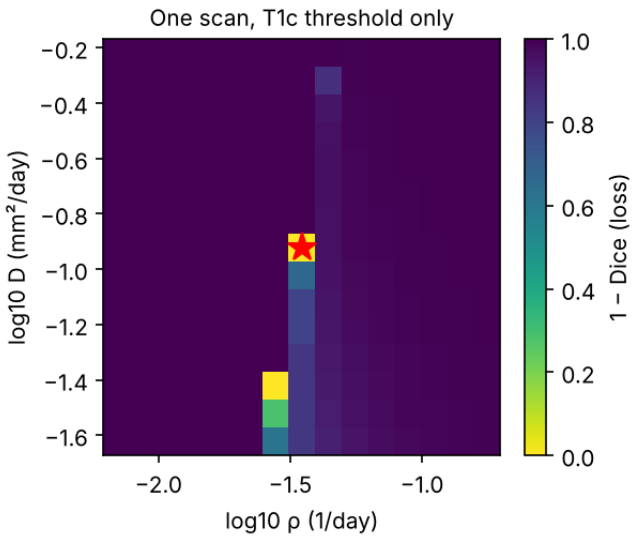
Worked Example: Is Tumour Growth Identifiable?

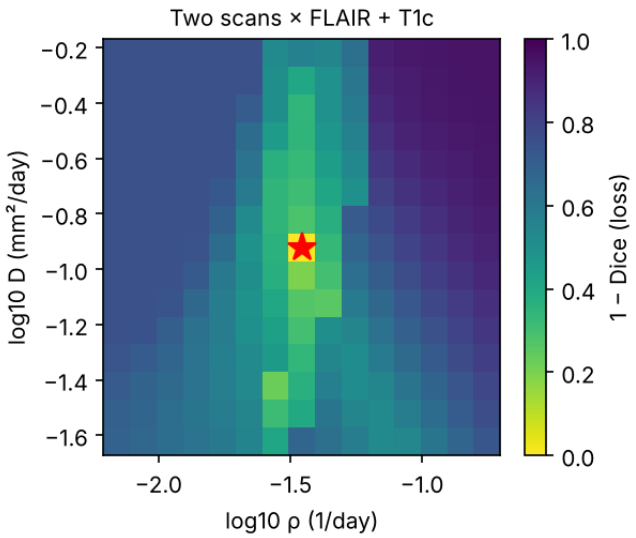
One scan admits different parameter sets; richer data separate them

$$\partial c/\partial t = \nabla\cdot(D\,\nabla c) + \rho\, c\,(1-c) \qquad \text{FLAIR visible: } c > 0.16 \qquad \text{T1c visible: } c > 0.8$$



Day 150 density. Yellow: T1c edge · cyan: FLAIR edge · dashed: c = 0.01





Assumptions

- Known seed and onset
- Known thresholds
- Isotropic D, no treatment

One T1c scan: D = 0.035, ρ = 0.027 fits as perfectly as the truth (D = 0.12, ρ = 0.035). Two scans × two contrasts: unique optimum at the truth. Predicted infiltration beyond the FLAIR edge (1457 vs 681 mm²) is a testable hypothesis.

Computed here: 2D Fisher–KPP model, 1 mm grid, assumed thresholds. Model family: Murray, Mathematical Biology (textbook).

Causal and Counterfactual Imaging AI



Ask what changes the prediction — and whether the change itself is realistic

Association vs causation

- Outputs correlate with pathology — or scanner, site, markers
- Causal claims need an explicit graph and assumptions
- Saliency maps are attributions, not causes

Counterfactual test

- Intervene on a suspicious region
- Observe change in tumour classification
- Large change ⇒ model relies on region (under assumptions)

Confounded interventions

- Black-box occlusion = non-physiological input
- Change may reflect artefact sensitivity
- Generative inpainting adds generator bias



Confounders open non-causal paths from acquisition to prediction.

Castro, Walker & Glocker, Causality matters in medical imaging, Nat. Commun. 2020, doi:10.1038/s41467-020-17478-w; Zech et al., PLoS Med. 2018.

Acquisition as Adaptive Experimental Design

Choose the next measurement to maximise expected information — within physics and hardware limits

$$\xi^* = \operatorname{argmax}_{\xi} \mathrm{EIG}(\xi), \quad \mathrm{EIG}(\xi) = \mathrm{E}_{p(y|\xi)} \left[\mathrm{KL}\left(p(\theta \mid y, \xi) \,\|\, p(\theta) \right) \right]$$

ξ	design: k-space lines, TIs, contrast injection, rescan timing
θ	quantity of interest: image, T1, lesion presence, growth parameters
y	measurement the design produces

Design problems already met: T1 inversion times, the second tumour scan — and, at hardware level, the 0.05 T scanner (Science 2024).

Adaptive k-space

- Sample where uncertainty is highest
- Learned masks (LOUPE), RL policies

Parameter-mapping design

- Choose TIs to minimise Cramér–Rao bound
- Module I: CSF limited by design

Longitudinal design

- When to rescan, which contrast
- Growth model: 2nd scan ⇒ identifiable

Across science

- Self-driving labs, active learning
- Optimal design in physics experiments

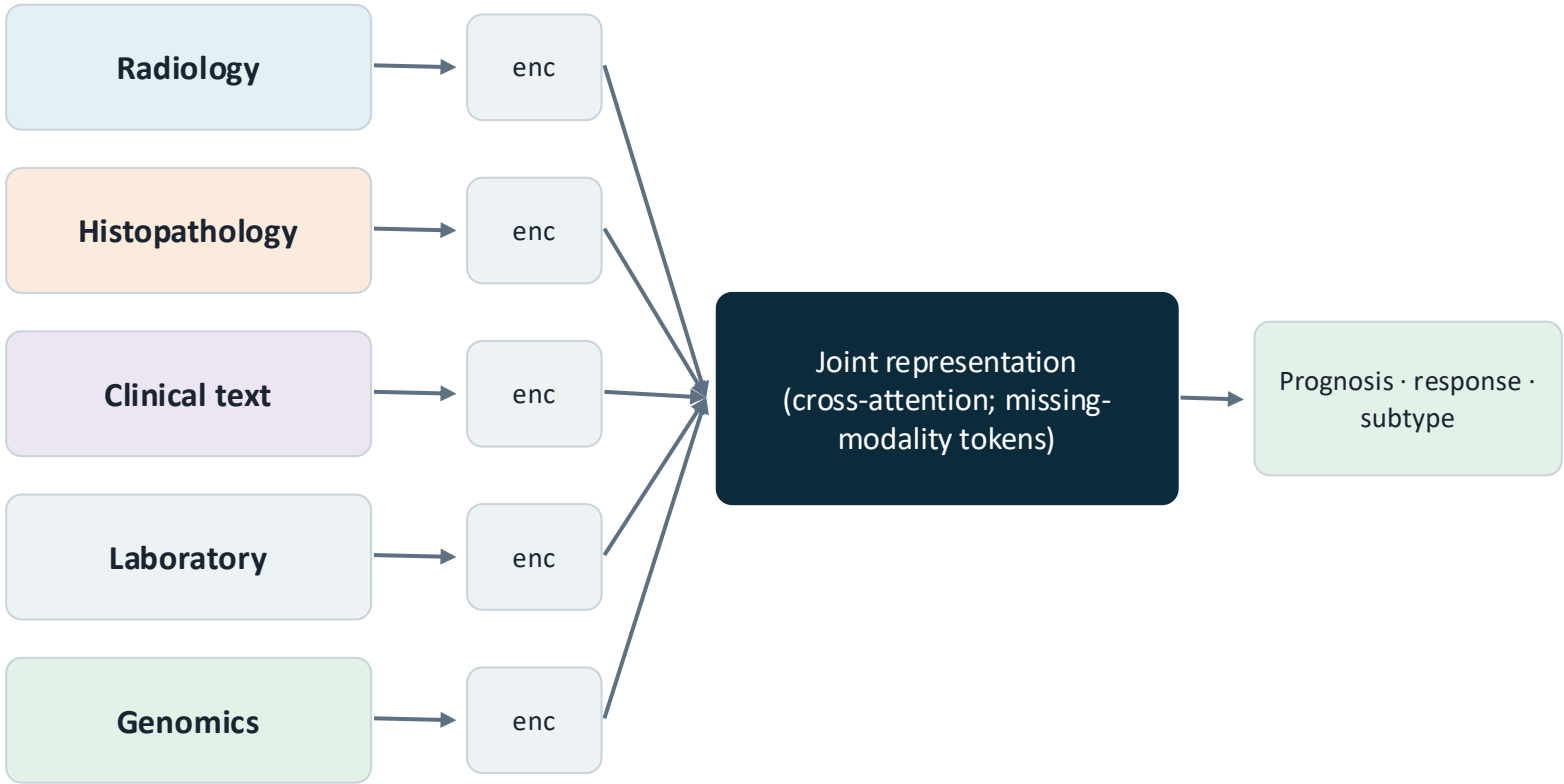
Constraints: k-space samples are global (never skip data consistency) · gradient slew and SAR limits · a task-optimised protocol can lose incidental findings.

Chaloner & Verdinelli, Bayesian experimental design: a review, Stat. Sci. 1995; Zhang et al., CVPR 2019; Bahadir et al., LOUPE, IEEE TCI 2020; Bakker et al., NeurIPS 2020.

Multimodal Precision Medicine

Diagnose · Decide

Fuse radiology, pathology, text, labs, and genomics — while handling what is missing



Fusion choices

- Early: needs complete aligned data
- Intermediate: most common
- Late: robust to missing, fewer interactions

Missing modalities

- Modality dropout, missing tokens
- Imputation inherits synthesis risks

Federated learning

- Share parameters, not data
- Heterogeneity; residual privacy risk

Lipkova et al., Artificial intelligence for multimodal data integration in oncology, Cancer Cell 2022; Pati et al., Federated learning enables big data for rare cancer boundary detection, Nat. Commun. 2022; Rieke et al., npj Digit. Med. 2020.

A Ladder of Scientific Claims

AI makes the lower rungs cheap — not the upper ones

1 Pattern

Held-out prediction. e.g. RETFound systemic disease

2 Robust association

Multi-site replication. e.g. PANDA

3 Measurement-grade

Bias, repeatability. e.g. CT-FM stability

4 Mechanistic hypothesis

Biological correlates. e.g. CHIEF image → genome

5 Interventional evidence

Prospective study. e.g. MASAI, PRAIM

6 Accepted knowledge

Independent replication, consensus — few AI examples yet

Where AI helps most

- Rung 1: finding candidate signals
- Rung 3: fast, precise estimation
- Design of the next experiment

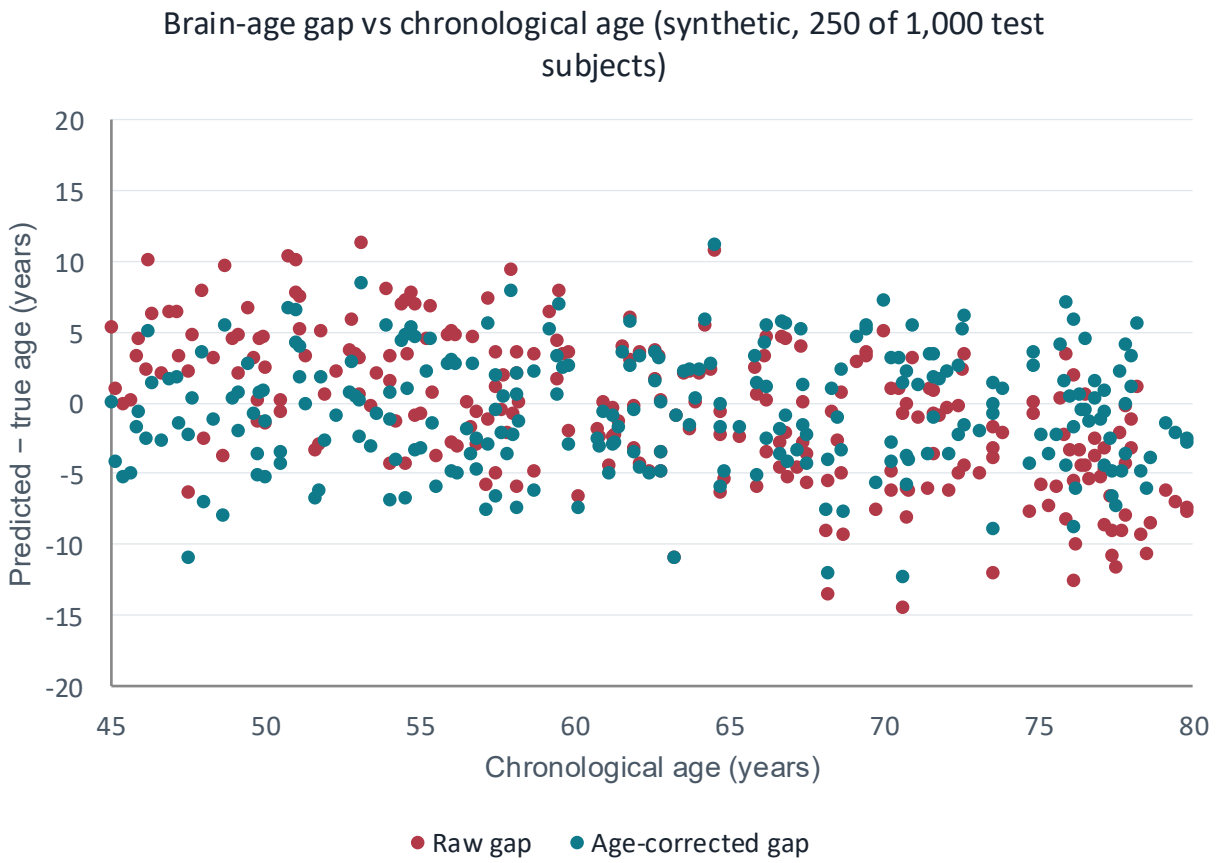
Where AI cannot substitute

- Rungs 4–6: mechanism, intervention, replication
- Most published AI ‘discoveries’ stop at rung 1–2

Wang et al., Scientific discovery in the age of artificial intelligence, Nature 2023; Ioannidis, Why most published research findings are false, PLoS Med. 2005.

Worked Example: The Brain-Age Gap

A learned image-derived phenotype — and a statistical artefact to remove before interpreting it



Computed (synthetic)	Raw gap	Corrected
Correlation with age	-0.58	0.02
Correlation with true deviation δ	0.56	0.68

The artefact

- Shrinkage toward the mean ⇒ gap falls with age
- Naive ‘discovery’: ageing slows in later life

Retinal photographs → CV risk factors

- Unexpected information exists (rungs 1–2)
- Mechanism and clinical value: open

Synthetic data computed here. Cole & Franke, Trends Neurosci. 2017; Smith et al., Estimation of brain age delta from brain imaging, NeuroImage 2019; Poplin et al., Prediction of cardiovascular risk factors from retinal fundus photographs, Nat. Biomed. Eng. 2018.

Discovery Pitfalls, Computed

With enough features, chance produces 'findings' — and circular evaluation certifies them

Simulation (no true signal)	Result
Features with p < 0.05 (of 1000)	56 (≈ 50 expected)
After Bonferroni	0
After Benjamini–Hochberg 5% FDR	0
AUC: select top-10 on all data, test on same	0.93
AUC: selection inside 5-fold CV	0.44

An 'excellent biomarker' (AUC 0.93) built from pure noise — certified by circular evaluation.

Forking paths

- Many analyses tried, few reported
- Agents can multiply this silently

Confounding

- Scanner, site, treatment drive features
- Stratify; negative controls

Remedies

- Pre-register; lock the analysis
- Nested validation; multiplicity control
- Independent replication

Computed here (simulated null data). Ioannidis, PLoS Med. 2005; Gelman & Loken, The garden of forking paths, Am. Sci. 2014.

Agents for the Scientific Process

The same architecture — specialists, evidence store, verifier — can run a study as well as write a report

Clinical agent (this lecture)	Research analogue	Key safeguard
Image quality	Data curation: cohort, inclusion criteria, site/scanner metadata	Documented selection; leakage checks
Segmentation + specialists	Measurement: validated pipelines	Bias and repeatability characterised
Cross-sequence fusion	Confounder and bias auditor	Site/scanner stratification; negative controls
Reporting	Analysis and write-up from a pre-registered plan	Locked plan; reproducible code
Verification	Statistical and reproducibility verifier	Re-runs; multiplicity; nested validation
Orchestrator	Research planner: hypothesis queue, next experiment	Provenance; every tried analysis logged

Human role

- PI approves hypotheses
- Domain experts interpret
- Responsibility stays human

New risk

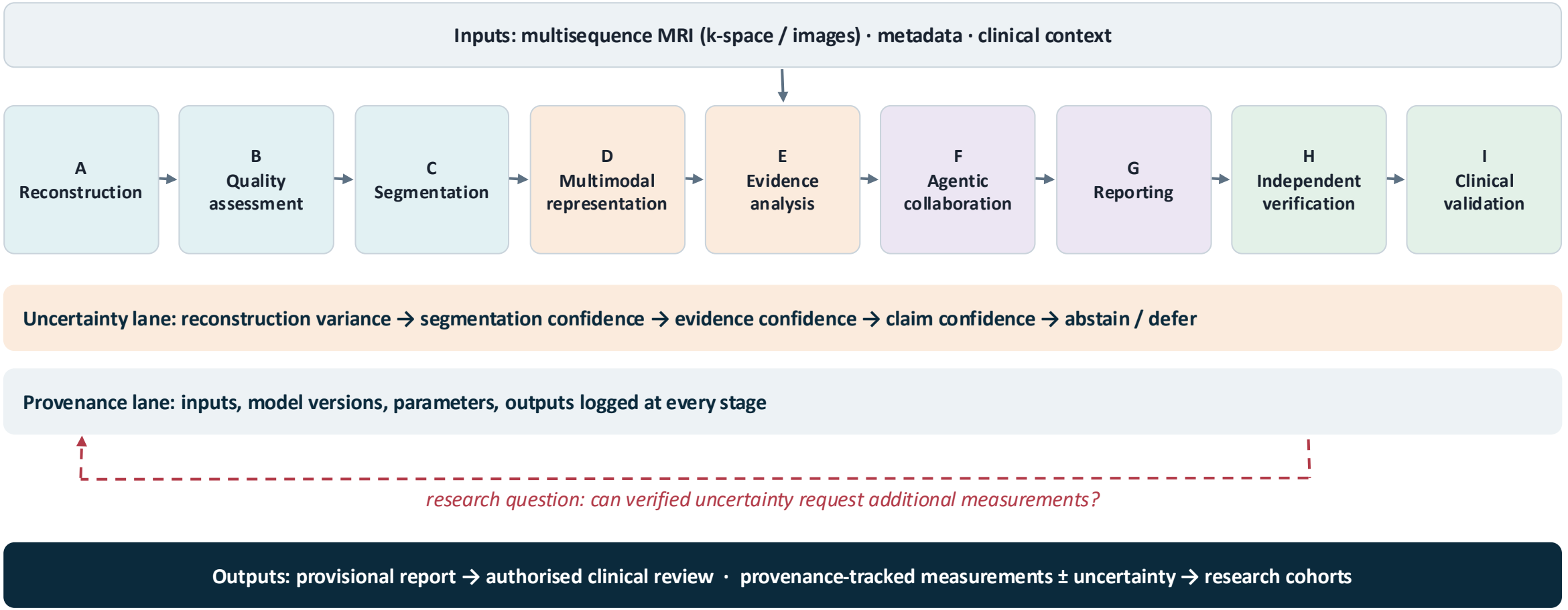
- Hypotheses at machine speed ⇒ multiplicity explodes
- Log every analysis tried

Lu et al., The AI Scientist, arXiv:2408.06292 (2024, preprint); Wang et al., Nature 2023. Mapping below is conceptual.

Case Study 6: Acquisition to Verified Report

End-to-end pipeline

The integrative research system — nine stages with explicit evidence and uncertainty flowing between them



Conceptual research architecture integrating Modules I–III; components draw on refs [nnU-Net, MedSAM2, RadFM, BrainGPT, verification literature].

Case Study 6: Stages A–E



Physics to evidence: what each stage consumes, produces, and how it is evaluated

Stage	Consumes	Produces	Evaluation
A Reconstruction	k-space, coil maps, mask	Images + uncertainty map	Lesion detectability, downstream Dice, reader study
B Quality assessment	Images, headers	Per-sequence QC scores, flags	Sensitivity to diagnosis-changing artefacts
C Segmentation	QC-approved sequences	Masks, volumes, voxel/lesion uncertainty	Dice, HD95, lesion-wise F1, volume agreement
D Multimodal representation	Image features, metadata, context	Joint embedding	Downstream tasks; cross-site robustness
E Evidence analysis	Masks, features, embedding	Observations {id, location, measurement, confidence}	Finding-level agreement with radiologists

Interfaces are typed and logged — each stage can be replaced or audited independently.

Case Study 6: Stages F–I

Diagnose · Report · Decide

Collaboration, reporting, verification, and the validation that would justify any of it

Stage	Consumes	Produces	Evaluation
F Agentic collaboration	Specialist observations	Reconciled evidence; escalations	Conflict resolution vs expert adjudication
G Reporting	Evidence store, context	Draft structured report citing IDs	Clinical entity / RadGraph F1; error taxonomy
H Verification	Draft + evidence	Approve / revise / defer / abstain	Recall on seeded errors; false-alarm rate
I Clinical validation	Provisional outputs	Evidence of clinical value	External consecutive cohort; reader study; prospective

Failure: safe but useless

Abstains on most hard cases

Failure: verified but wrong

Consistent report built on wrong evidence

Failure: too slow

Latency incompatible with emergency workflow

CLAIM 2024 (doi:10.1148/ryai.240300); TRIPOD-LLM (doi:10.1038/s41591-024-03425-5).

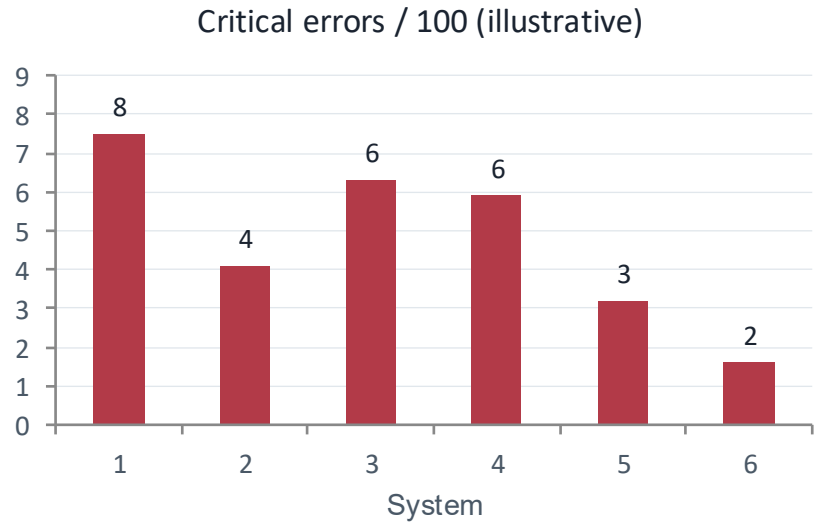
Case Study 6: Ablation Design

Each added component must earn its latency and complexity

ILLUSTRATIVE VALUES — not experimental results

System	Finding F1	Critical errors / 100	Unsupported stmts	Abstain / defer	Latency (s)
1 Single CNN (+ template)	0.61	7.5	— (template)	0%	2
2 CNN + segmentation	0.70	4.1	— (template)	0%	6
3 Multimodal foundation model	0.72	6.3	9.8%	0%	12
4 FM + retrieval	0.75	5.9	7.1%	0%	15
5 Multi-agent system	0.78	3.2	5.4%	4%	70
6 Multi-agent + verification	0.79	1.6	1.2%	9%	95

Is −1.6 critical errors / 100 worth +25 s per study and 9% deferrals? Elective outpatient MRI ≠ emergency stroke CT.



Reading rules

- Abstention changes the denominator: compare at matched coverage
- Paired tests with CIs, external cohort

CNN · ViT · Foundation Model · VLM · Agents

Cross-cutting

Maturity decreases as flexibility increases

	Traditional CNN	Vision Transformer	Medical foundation model	Multimodal VLM	Agentic AI
Required data	Hundreds of labelled cases	More labels or pretraining	Large unlabelled corpus + small labelled	Paired image–report data	Component data + policy / interaction validation
Representation	Local, hierarchical	Global token interactions	Transferable general features	Aligned image–text space	Composed from tools and agents
Adaptability	Retrain per task	Fine-tune	Prompt, PEFT, fine-tune	Prompt, instruction tuning	Dynamic tool selection
Compute	Low–moderate	Moderate–high (N^2)	High pretraining	High (LLM inference)	Multiplied inference calls
Interpretability	Saliency (limited)	Attention maps (limited)	Limited	Text explanations (may be unfaithful)	Logs and evidence trails; complex
Failure modes	Shift, shortcuts	Data hunger, shift	Small/low-contrast targets	Hallucination, omission	Propagation, correlated errors
Clinical maturity	Regulated products exist	Research → some products	Research, early translation	Research; reader studies	Mostly conceptual / prototypes

Limitations, Safety, and Accountability

Diagnose · Report · Decide

From prediction to diagnosis there are four distinct steps — never collapse them

Model prediction → **Model explanation** → **Image-derived evidence** → **Verified clinical finding** → **Final clinical diagnosis (accountable clinician)**

Data

- Dataset bias; rare pathology
- Annotation disagreement
- Patient-level leakage

Shift

- Scanner and protocol shift
- Inter-institutional generalisation
- Missing sequences

Model outputs

- Small-object segmentation
- Diagnostic hallucinations
- Unsupported report findings
- Calibration failure

Use

- Automation bias
- Privacy; synthetic-data reliability
- Prospective validation lacking
- Human accountability

Confidence ≠ calibrated probability · agents are not independent · agreement ≠ correctness · every generated report is a provisional research output requiring authorised human review.

Dratsch et al., Automation bias in mammography, Radiology 2023, doi:10.1148/radiol.222176; CLAIM 2024; TRIPOD-LLM 2025.

Open Research Questions

Cross-cutting

Each is a plausible master's or doctoral project

1. How can diagnostic evidence be preserved during AI reconstruction?
2. Can foundation models learn robust volumetric representations with limited labelled data?
3. How can image and clinical evidence be combined without producing unsupported claims?
4. When does multi-agent collaboration actually improve performance?
5. How should agent disagreement be quantified?
6. How can an AI system learn when it should abstain?
7. Can uncertainty guide acquisition, reconstruction, segmentation, and reporting jointly?
8. How should longitudinal clinical imaging be modelled?
9. What is required to move from retrospective validation to prospective deployment?
10. How can AI produce genuinely new scientific insight rather than statistical correlation?
11. When is an AI-derived image feature a measurement — with known bias and repeatability?
12. How can learned and mechanistic models be combined so that fitted parameters remain identifiable?

Five Take-Home Messages

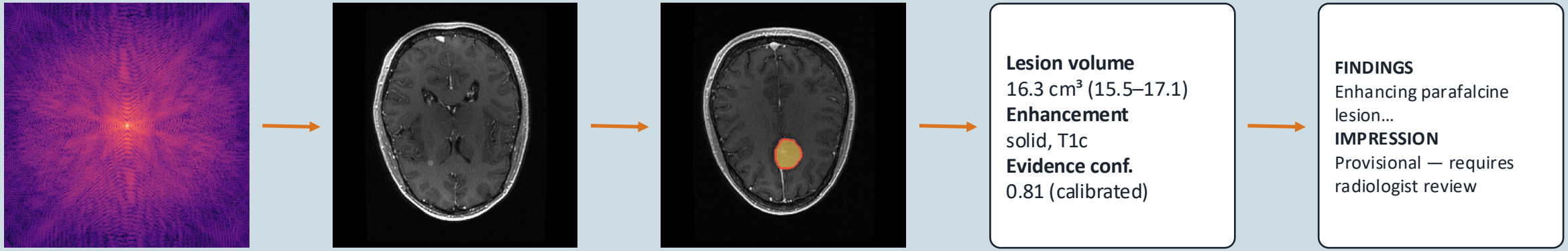
1. Medical imaging begins with physical measurements and constrained inverse problems.
2. Deep learning provides powerful representations for quantitative image analysis.
3. Foundation models and multimodal learning support reusable, flexible medical intelligence.
4. Agentic AI can coordinate specialized models but also introduces new error propagation and governance risks.
5. Reliable medical AI requires calibrated uncertainty, evidence verification, robust validation, and clinical oversight.

Can the next generation of medical AI move beyond recognizing image patterns to producing reliable, verifiable, and scientifically meaningful clinical knowledge?

AI for Medical Imaging Science

From Physical Measurement to Clinical and Scientific Knowledge

Essam Rashed, Ph.D.
Graduate School of Information Science,
University of Hyogo



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[J] journal · [C] conference · [P] preprint · [R] review · [G] guideline · [B] book · [D] data

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Discussion Questions: Foundational

Cross-cutting

Check understanding of physics, metrics, and validation

F1. Why is reconstruction from 4× undersampled Cartesian k-space ill-posed? What role does the regulariser play?

F2. Derive $p = -\ln(I/I_0)$ from Beer–Lambert. If dose is halved, how does image noise change?

F3. $|P| = 50$, $|G| = 40$, $|P \cap G| = 30$ voxels. Compute Dice and IoU.

F4. 25,000 tumour voxels: what volume at 1×1×1 mm^3 and at 0.9×0.9×5 mm^3? Why read spacing from the header?

F5. Distinguish internal held-out testing, external validation, and prospective evaluation. What claim does each support?

Discussion Questions: Advanced Technical

Derivations and calculations from Modules I–III

A1. Show that IoU = Dice / (2 − Dice).

A2. Why does self-attention divide QK^T by $\sqrt{d_k}$? State the assumption.

A3. Derive $q(x_t \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\, x_0, (1 - \bar{\alpha}_t)I)$ from the one-step forward process.

A4. Test–retest T1 scans give within-subject SD σ_w = 30 ms. Compute the repeatability coefficient. Is a 60 ms change real?

A5. An ensemble predicts p(lesion) = 0.95, 0.05, 0.50. Compute total entropy, expected entropy, and mutual information (nats). Interpret.

Discussion Questions: Research-Oriented

Cross-cutting

Open-ended — answers are designs, not numbers

R1. Design an experiment testing whether a learned MRI reconstruction preserves small lesions.

R2. How would you measure whether two agents' errors are independent?

R3. Propose an evaluation protocol for a report-verification agent.

R4. What evidence would justify prospective deployment of a report generator?

R5. How would you distinguish a discovered imaging biomarker from a scanner or site confound?

Research Assignment

Design an Evidence-Grounded AI System for Medical Image Interpretation

1 Research problem

Clinical need and scientific question

2 Dataset

Source, size, labels, licence, splits by patient/site

3 System architecture

Image model + foundation/multimodal model; evidence interface

4 Scientific hypothesis

Falsifiable; outputs characterised as measurements (bias, repeatability)

5 Evaluation methodology

Task metrics, clinical metrics, external test, reader study design

6 Comparison baselines

nnU-Net or equivalent; VLM without evidence; ablations

7 Uncertainty and validation

Calibration, abstention, OOD, error taxonomy, CIs

8 Expected contributions

New knowledge or capability; rung on the ladder of claims reached

Assignment Deliverables and Assessment

Cross-cutting

Graded on scientific rigour, evidence design, and honesty about limits

Criterion	Weight	What earns full marks
Problem & hypothesis	15%	Clinically motivated, falsifiable, primary endpoint stated
Data & splitting	15%	Licensed data; patient- and site-level splits; leakage addressed
Architecture & evidence path	20%	Every output claim traceable to image-derived evidence
Evaluation & baselines	20%	Task + clinical metrics; strong baselines; ablations; CIs
Uncertainty & measurement	15%	Calibration, abstention, OOD; bias and repeatability of outputs
Limitations & contribution	10%	Honest scope; what remains unproven
Clarity & figure	5%	One clear architecture figure; reproducible description

Deliverables

- 6–8 page proposal
- One architecture figure
- Evaluation plan table
- 1-page risk & limitation register

Common pitfalls

- BLEU/PSNR as sole metric
- Weak baselines
- Agreement treated as validation
- Readiness claimed from benchmarks